steam rate is 0.25 mole per mole of the feed. At this low concentration, the equilibrium relation is approximately linear, $y = 8.5x$. Determine the number of ideal trays required if 95% of the ethanol is to be removed.

**7.55** ***(A column with side streams)***[3] A distillation column should separate 200 kmol/h of a saturated liquid feed containing 45 mole% methanol and 55 mole%. The top product should have 97 mole% methanol and the bottom, 96 mole% water. The reflux is saturated liquid and a reflux ratio of 2.0 is suggested. A vapour side stream containing 80 mole% methanol is to be withdrawn at a rate of 50 kmol/h from the appropriate tray and condensed in a heat exchanger. Half of it should be taken as an intermediate product and the rest returned into the column as an additional reflux stream. Determine (a) the number of ideal stages required, (b) the feed tray, (c) the tray from which the side stream should be withdrawn, and (d) the tray that should receive the additional reflux.

## REFERENCES


Billet, R., *Distillation Engineering*, Chemical Publishing, 1979.

Billet, R., *Packed Towers in Processing and Environmental Engineering*, VCH, 1995.

Divekar, U.M., *Batch Distillation*, Taylor and Francis, 1995.

Eubank, P.T. et al., 'Simplify flash calculations', *Chem. Eng.*, Nov. 2000, 125–130.

Fair, J.R., 'Distillation, whither, not whether', *Chem. Eng. Res. & Des.*, *66*(July 1988) 363–370.

Forbes, R.J., *Short History of the Art of Distillation*, E.J. Brill, Leiden, 1948.

Hala, M. et al., *Computer-aided Data Book of Vapor-Liquid Equilibria*, Kodansa, Tokyo, 1975.

Jafarey, A., J.M. Douglas, and T.J. McAvoy, 'Shortcut techniques for distillation column design and control column design', *Ind Eng. Chem. Proc. Des. Dev.*, *18*(1979) 197–202.

Kister, H.Z., *Distillation Operation*, McGraw-Hill, New York, 1989.

Kister, H.Z., *Distillation Design*, McGraw-Hill, New York, 1992.

Kunesh, J.G. et al., 'Distillation: still towering over other options', *Chem. Eng. Progr.*, Oct. 1995, 43–54.

Lewis, W.K., 'The efficiency and design of rectifying columns for binary mixtures,' *Ind. Eng. Chem.*, *14*(1922) 492.

Porter, K.E., 'Why research is needed in distillation', *Trans. Inst. Chem. Engrs.*, *73* (Part A), May 1995, 357–362.

Ryan, J.M., 'Replotting the McCabe-Thiele diagram', *Chem. Eng.*, May 2001, 109–113.

Smoker, E.H., 'Analytic determination of plates in fractionating columns', *Trans. AIChE*, *34*(1938) 165–172.

Sorel, E., *La Rectification de l'Alcohol*, Gauthiers Villars, Paris, 1893.

Stichlmair, J.G., and J.R. Fair, *Distillation—Principles and Practice*, Wiley-VCH, New York, 1998.

Yuan, K.S., and B.C.-Y. Lu, 'Vapour-liquid equilibria,' *J. Chem. Eng. Data*, *8*(Oct. 1963) 549.

# 8 Liquid–Liquid Extraction

Liquid–liquid extraction (sometimes abbreviated LLX) is a mass transfer operation in which a solution (called the *feed* which is a mixture of a solute and a carrier liquid) is brought into intimate contact with a second immiscible or slightly miscible liquid (called the *solvent*) in order to achieve transfer of the solute(s) from the feed to the solvent. The two liquid phases that have different densities are then separated. The solute-rich phase (this is the solvent stream, now enriched with the solute) is called the *extract*; the residual feed stream that may have a little of the solute left in it is called the *raffinate*. If the carrier in the feed and the solvent are partly miscible, the raffinate will also have a little solvent dissolved in it, and the extract phase will contain a little dissolved carrier. The solute is separated from the extract phase in an acceptably pure form, and the solvent dissolved in the raffinate is recovered. An extraction process generally involves four major steps.

(a) Bringing the feed and the solvent into intimate contact by dispersing one phase into the other as droplets.
(b) Separation of the extract and the raffinate phases that have different densities.
(c) Removal and recovery of the solute from the extract phase in a relatively pure form (by distillation, evaporation, crystallization, etc.).
(d) Removal and recovery of the solvent from each phase, usually by distillation.

Separation of a liquid mixture or solution by distillation is operationally simpler than extraction. For example, a binary liquid mixture having components of different volatilities can, in principle, be separated into two relatively pure products (a 'distillate' and a 'residue') in a single column. But separation of the same mixture by extraction requires *at least* one distillation column in addition to the extractor. Very often more than one distillation column is used for the recovery of the solvent from the raffinate (called *raffinate clean-up*) and for separation of the carrier liquid from the extract phase (Figure 8.1). Phase separation of the extract and the raffinate may not be rapid if the difference in their densities is small and the interfacial tension is low. Loss of solvent may also occur in some cases. In its totality, solvent extraction is a more elaborate technique than distillation. In a typical liquid–liquid extraction process, only about 15% of the capital cost goes towards the extractor and the remaining 85% goes towards the distillation

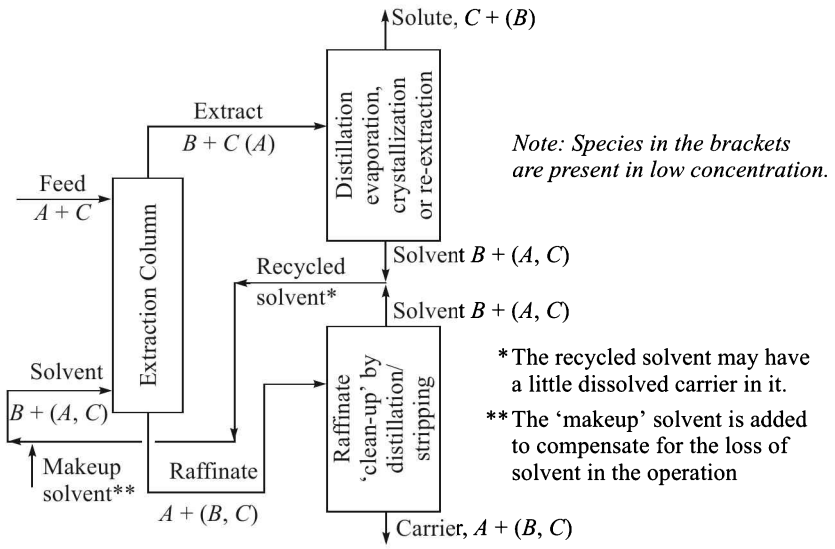


**Figure 8.1** Schematic of an extraction unit with product recovery and raffinate clean-up arrangements; *A*: carrier, *B*: solvent, *C*: solute.

columns and other equipment for solvent recovery and product purification. Also, only about 5% of the operating cost of the unit is required for the extractor and the remaining 95% goes towards solvent recovery. Despite these limitations, there are situations when the technique of distillation either fails or becomes uneconomic. A few such cases, where solvent extraction is an attractive separation technique, are described below.

(a) The components to be separated have close boiling points. For example, aromatics are extracted from paraffinic feedstocks in the refineries by using a solvent like sulphur dioxide, *n*-methyl pyrolidone, sulpholane, etc. The number of distillation trays becomes prohibitively large if the relative volatility is close to unity (say less than 1.2), and solvent extraction, therefore, may be an attractive alternative for such a mixture.
(b) Separation of heat-sensitive materials such as antibiotics, vitamins, etc. Penicillin is recovered from the fermentation broth by using a suitable solvent such as butyl acetate.
(c) Recovery of non-volatile solutes, usually from aqueous solutions, in hydro-metallurgy.
(d) Recovery of a solute from a very dilute solution—for example, bioseparation (proteins, antibiotics, high-value bioproducts).
(e) Removal of organics from aqueous streams—for example, removal of phenol from aqueous wastes.

## 8.1 A FEW EXAMPLES OF SOLVENT EXTRACTION

Liquid–liquid extraction is also called 'solvent extraction'. A number of common industrial extraction processes are listed in Table 8.1 (also see Cusack and Glatz, 1996). A few such representative processes are briefly described below.

**Table 8.1** Some commercially important extraction processes

| | *Extraction process* | *Feed* | *Solvent* |
|---|---|---|---|
| 1. | Separation of BTX (benzene, toluene and xylenes) from petroleum fractions. (BTX is used as a feed-stock for the preparation of petrochemicals). | Petroleum fractions containing aromatics | Sulpholane (tetrahydrothiophene, 1-1-dioxide), furfural, N-methylpyrrolidone, diethylene glycol, tetraethylene glycol. |
| 2. | Lube oil extraction | Hydrocarbons including *n*-paraffins, *iso*-paraffins, naphthenes, aromatics and mixed aliphatic-aromatic ring structures | Liquid $SO_2$, phenol, furfural, liquid propane, N-methyl-pyrrolidone |
| 3. | Separation of aromatic hydrocarbons from petroleum fractions | Petroleum fractions containing aromatics | Tetraethylene glycol–water |
| 4. | Extraction of caprolactum from 'lactum oil' | 'Lactum oil', aqueous $(NH_4)_2SO_4$–liquor containing caprolactum (such liquids are obtained in the process of caprolactum manufacture) | Benzene, toluene, trichloroethylene, etc. |
| 5. | Extraction of phenolics from coal tar and liquors | Light liquor containing phenolics (obtained in a coal carbonization plant) | Isobutyl acetate |
| 6. | Extraction of phenol from aqueous wastes | Aqueous waste streams, coal carbonization plants, phenolic resin plants | Butyl acetate, di-isopropyl ether, MIBK |
| 7. | Extraction of citral (a flavouring substance used in food and some other industries) | Lemon grass and orange oil | An alcohol and a lower hydrocarbon(solvent pair; for example, methanol and pentane) |
| 8. | Refining of oils and fats (mono- and diglycerides removed from tri-glycerides) | Oil and fat products | A polar solvent like acetone or a lower alcohol |
| 9. | Extraction of copper | Dilute Cu-ion bearing acidic leach liquor | Hydroxylamine, proprietary solvents like LIX-63, LIX-64N, LIX-65N, LIX-70, etc. |
| 10. | Extraction of penicillin | Penicillin fermentation broth | *n*-Amylacetate (most common), *n*-butyl acetate |
| 11. | Removal of $H_2S$ | Liquefied natural gas (LNG) | Monoethanolamine (MEA), methanol-diethanolamine amine (MDEA). |

### 8.1.1 Extraction of Aromatics

Solvent extraction is an important separation technique in petroleum refineries, particularly for lube oil production. The viscosity index (V.I.) is an important parameter for a lubricating oil. Many undesirable compounds, especially aromatics that reduce the V.I., have to be removed from the lube oil feed stock to ensure the required viscosity index. Aromatics (benzene, toluene and xylenes, BTX) must also be removed from kerosene and jet fuels. This can be done conveniently by using a solvent selective for aromatics, sulphur, nitrogen, and oxygenated compounds. Important solvents are furfural, sulpholane (tetrahydrothiophene-1,1-dioxide), N-methyl pyrrolidone (NMP), diethylene glycol, tetraethylene glycol, methyl-ethyl-ketone (MEK), and methyl-isobutyl-ketone (MIBK) which selectively remove aromatics and weakly polar compounds (Lucan, 2000). Phenol and sulphur dioxide are no longer used because of their toxicity and other problems. The process involves extraction followed by distillation-stripping of both the raffinate (that contains the purified lube and some solvent) and the extract (containing the aromatics and the solvent). A very simple flow diagram of the sulpholane process is shown in Figure 8.2(a). A rotating disk contactor (RDC, see Section 8.5.4) is commonly used for this purpose. The separated aromatics are further fractionated in pure form for use as feedstocks for the petrochemical industry.

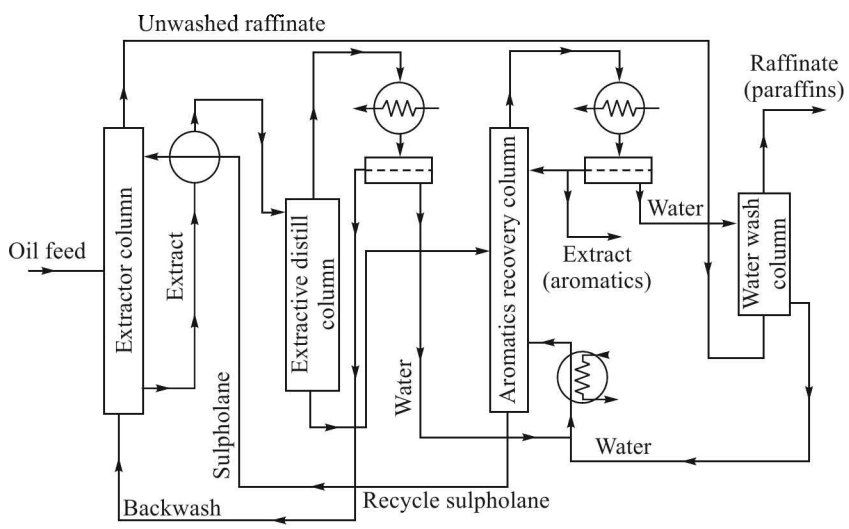


**Figure 8.2(a)** Separation of aromatics from a feedstock—the sulpholane process.

### 8.1.2 Recovery and Concentration of Antibiotics by Solvent Extraction

Penicillin, a major antibiotic, is produced by fermentation. The concentration of the substance in the fermentation broth is small (typically 20–35 gram penicillin per litre broth). Recovery and concentration of penicillin from the broth is done by extraction (Hatton, 1985) using a solvent such as amyl acetate, butyl acetate, or methyl-isobutyl-ketone (MIBK). At a pH of about 2, penicillin remains in the acid form and has a low solubility in water. But at this pH, it is favourably

transported to an organic solvent (mentioned earlier) from the aqueous fermentation broth because of a high distribution coefficient. However, a reversal of the distribution coefficient occurs at a pH above 7. This pH-dependent distribution of penicillin in the water-ester pair is made use of to recover, purify and simultaneously concentrate the antibiotic from the broth. The extraction process, which is carried out at about 4°C, is schematically shown in Figure 8.2(b).

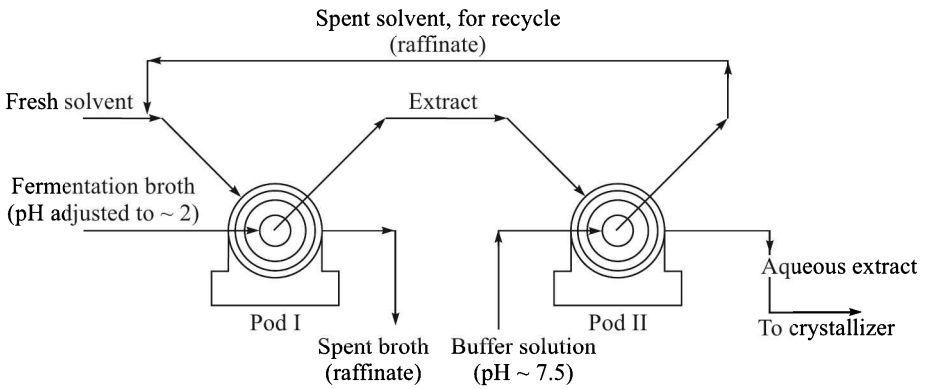


**Figure 8.2(b)** Recovery and concentration of penicillin by extraction.

The pH of the fermentation broth is adjusted to about 2 by dosing sulphuric acid. It is then brought into intimate contact with the solvent in a centrifugal extractor (Pod I). A Podbielniak extractor (often abbreviated 'Pod') is commonly used and a solvent-to-broth ratio of 0.1 to 0.2 is maintained. More than 90% recovery of penicillin occurs and its concentration in the solvent or ester-phase becomes a few times more than that in the feed broth. The penicillin-rich extract (the ester-phase) is now contacted with a basic buffer solution of pH 7.5 in a second extractor (Pod II) when most of the antibiotics get transferred into the aqueous phase in the salt form (generally it is *potassium penicillin*). The concentration in the extract further increases a few times in this extractor that uses an organic-to-aqueous phase ratio of 5 to 10. The raffinate from the second extractor (the ester-phase) is recycled to Pod I as the solvent. Penicillin is crystallized from the aqueous extract by adjusting the pH once again.

### 8.1.3 Extraction of Caprolactum

Caprolactum is produced and used in large quantities for the manufacture of nylon-6, a bulk polymer. It is synthesized from cyclohexane (cyclohexane → cyclohexanol → cyclohexanone → cyclohexanone oxime → caprolactum). The last step of the process involves reaction of the oxime with sulphuric acid when caprolactum and ammonium sulphate are formed. The aqueous reaction mixture separates into an organic-rich phase ('lactum oil' containing 65–70% caprolactum) and a water-rich phase having ammonium sulphate and 2–3% caprolactum. The two phases are fed at appropriate locations of a rotating disk contactor [Figure 8.2(c)]. Toluene, benzene and trichloroethylene are good solvents (the last two are carcinogens and are not used any more). Caprolactum goes to the extract, and ammonium sulphate remains in the raffinate. The two streams are further treated to get caprolactum in a pure form and to recover the solvent and ammonium sulphate.

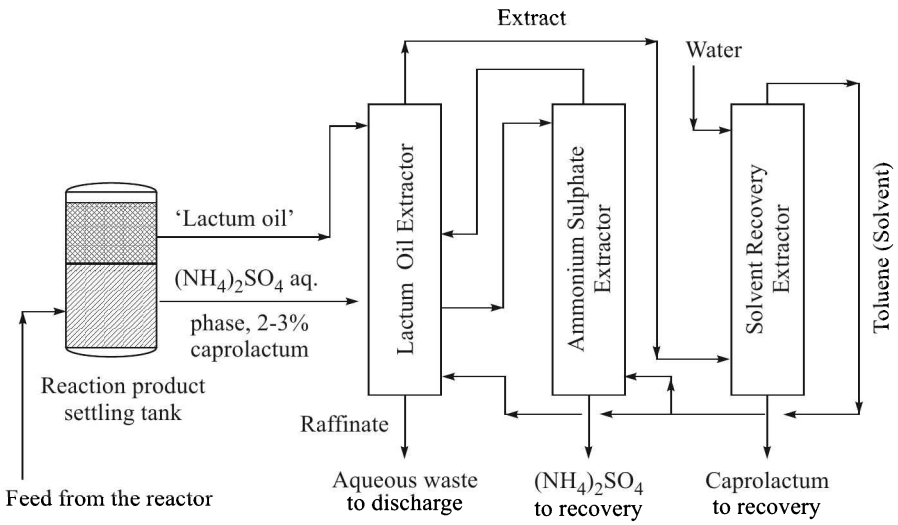


**Figure 8.2(c)** Recovery of caprolactum by solvent extraction.

### 8.1.4 A Few Other Applications

Solvent extraction has many other uses in food, pharmaceutical, metallurgical, environmental, and organic and inorganic chemical industries. It is used for purification of wet-process phosphoric acid to prepare the food-grade material. Recovery of acetic acid from a dilute aqueous solution has been practised since long because separation by distillation is very energy-intensive. Ethyl acetate and di-iso-propyl ether are the preferred solvents. Extraction of protein from ground fish or refining of fats by using propane as the purifying solvent (Solexol process) are typical applications in the food industry. Natural vitamins A and D are extracted from fish liver oil using liquid propane. Solvent extraction has proved to be effective in wastewater treatment. A typical application is separation and recovery of toxic organics like phenol. With the rapid growth in biotechnology, separation of high value bioproducts from dilute aqueous solutions (often called *bioseparation*) has assumed great importance. Solvent extraction is an effective and widely used technique for this purpose. One novel application is separation and concentration of proteins and other biomolecules by 'aqueous two-phase extraction' (Rao et al., 1998). The carrier and the solvent phases are aqueous solutions of two mutually incompatible polymers. The most common two polymers used for this purpose are polyethylene glycol (PEG) and dextran. Two major disadvantages of the technique occur due to small density difference and low interfacial tension of such two 'immiscible' aqueous phases.

Solvent extraction is an important technique of recovery of some metals. Extraction of copper from the ore leach liquor is a very important process. It can recover copper from very dilute solutions. A chelating extractant derived from hydroxyoximes or quinolinol in kerosene solution is largely used. Applications to nickel and cobalt extraction are also well known. Solvent extraction is an important separation step in 'nuclear fuel reprocessing'. Spent fuel from a nuclear reactor contains $^{238}U$, $^{235}U$, $^{239}Pu$, $^{239}Th$, and a few other fission products. Reprocessing consists of a number of steps for the separation of plutonium and residual uranium from the rest with a view to reusing them by mixing with fresh nuclear fuel. Alternatively, these materials may be disposed of safely.

The spent fuel is first dissolved in nitric acid, and the hexavalent basic nitrates of uranium and plutonium—$UO_2(NO_3)_2$ and $PuO_2(NO_3)_2$—are extracted from the aqueous solution using a solvent like tri-*n*-butyl phosphate mixed with kerosene. The metals are then recovered from the extract and recycled. Separation and concentration of uranium from the ore leach liquor are also done by solvent extraction. The mixer-settler type extractors are commonly used in metal extraction.

## 8.2 LIQUID–LIQUID EQUILIBRIA (LLE)

A liquid–liquid extraction system contains at least three components—the solute (*C*), the carrier liquid in the feed (*A*), and the extracting solvent (*B*). For example, in the process of extraction of penicillin from the fermentation broth described in Section 8.1, penicillin is the solute (*C*), water is the carrier (*A*), and butyl acetate is the solvent (*B*). Quite often the total number of components is more than three; the carrier itself may have more than one liquid in it, there may be more than one solute in the feed, or the solvent may be a mixed solvent. The equilibrium data for a liquid–liquid system are of vital importance in the selection and design of an extraction equipment.

### 8.2.1 Classification of Ternary Systems

Here we shall confine our discussion only to ternary systems comprising a carrier (*A*), a solvent (*B*), and a solute (*C*). Three binary mixtures can be formed out of these components: *A–B*, *B–C* and *C–A*. The mutual miscibility behaviour of the components in each of these binaries determines the nature of the equilibrium diagram for the ternary system. Most of the ternary systems fall in one of the following categories.

(a) The carrier (*A*) and the solvent (*B*) are practically immiscible. The equilibrium in such a system may be represented in the form of a '*x–y*' plot as in the case of distribution of a solute between a carrier gas and a solvent. If the solute concentration is small, a Henry's law-type linear distribution law applies.

(b) The solute is miscible with the carrier and with the solvent in all proportions, but the carrier and the solvent are only partially miscible (*A–C* and *B–C* are miscible pairs; the pair *A–B* is partially miscible). Such a system is called a Type-I ternary system. About 75% of ternary liquid systems fall in this category.

(c) The solute is completely miscible with the carrier; but both the solute and the carrier have limited miscibility with the solvent (the binary *A–C* is miscible in all proportions; the pairs *B–C* and *A–B* are only partially miscible). A ternary mixture of this type is called a Type II system.

### 8.2.2 Experimental Determination of Liquid–liquid Equilibrium Data

As in the case of determination of gas–liquid or vapour–liquid equilibria, liquid–liquid equilibrium data are obtained by measuring the concentrations of the three components in the two liquid phases in equilibrium at a given temperature. The following steps are followed: (a) Suitable

quantities of purified *A, B*, and *C* are taken in an equilibrium cell maintained at a constant temperature. The content is mixed vigorously for a sufficient time to ensure attainment of equilibrium. (b) The content is allowed to separate into two phases. (c) Samples are drawn from the two phases and analyzed for the concentrations (expressed as mass fractions or mole fractions) of *A, B*, and *C*. A set of such experiments conducted with varying amounts of the three components at a constant temperature yields the data for the construction of a liquid–liquid equilibrium diagram. The liquid phase rich in the solvent (*B*) is called the *extract*, and the other phase which is rich in the 'carrier' is called the *raffinate*.

### 8.2.3 Graphical Representation of Liquid–liquid Equilibrium Data

For any of the two liquid phases in equilibrium in a ternary system, there are three concentration terms of the three components. If concentrations are expressed in mole fraction or in mass fraction, the sum of the three concentrations is unity. It is not possible to describe a phase or the equilibrium of two phases in contact in terms of the three concentrations using the normal rectangular coordinate system. But this can be done conveniently on an 'equilateral triangular graph'. Also there are two other techniques of representation of ternary equilibrium data using the rectangular coordinate system: (a) a 'right-angled triangular diagram' in which only two concentrations are plotted, and (b) using rectangular coordinates and plotting the mass fractions (or mole fractions) of the solute and of the solvent in the two phases on 'solvent-free basis'.

#### *The equilateral triangular diagram*

The basis of the technique of presentation of equilibrium data on an equilateral triangular graph paper is that the sum of the distances of a point within an equilateral triangle from the three sides is equal to the height of the triangle. On the equilateral triangle *ABC* in Figure 8.3(a), the vertices *A, B* and *C* represent the 'pure components' *A, B* and *C* respectively; the sides *AB, BC* and *CA*

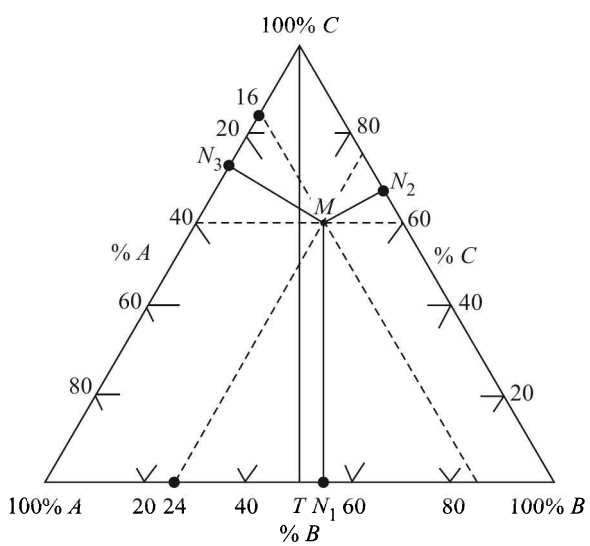


**Figure 8.3(a)** Liquid–liquid equilibrium data presented in the equilateral triangular coordinates.

represent mixtures of $A$ and $B$, $B$ and $C$, and $C$ and $A$ respectively. A point within the $\Delta ABC$ represents a mixture of the components $A$, $B$ and $C$. Consider the point $M$, for example. The sum of its distances from the three sides is $MN_1 + MN_2 + MN_3 = CT$, which is equal to the altitude of the triangle. From Figure 8.3(a) it is found that the point $M$ lies on the dotted line representing 60% $C$ ($MN_1 = 0.6CT$, $CT$ being the altitude). Also, the point $M$ lies on the dotted lines representing 16% $A$ and 24% $B$ respectively. So a mixture of $A$, $B$ and $C$ having 16% $A$, 24% $B$, and 60% $C$ is represented by the point $M$ on the triangular diagram. The LLE diagram of a ternary system is depicted in Figure 8.3(b). Here the solute $C$ is miscible with the carrier $A$ and with the solvent $B$ in all proportions, but the carrier and the solvent are only partially miscible (Type I system). The curve $RPS$ is the equilibrium diagram in the equilateral triangular coordinate system. It is obtained by plotting the data points representing the two liquid phases in equilibrium. The point $R$ on the line $AB$ represents the solubility of the solvent $B$ in the carrier $A$; the point $S$ represents that of the carrier $A$ in the solvent $B$ if there is no solute.

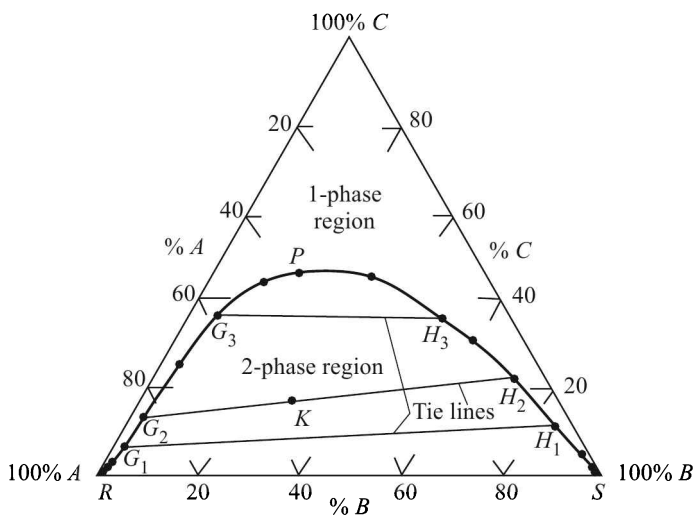


**Figure 8.3(b)** Liquid–liquid equilibrium data presented in the equilateral triangular coordinates.

The pair of points $G_1$ and $H_1$ on the curve represents two liquid phases in equilibrium; $G_1$ is rich in the carrier (*raffinate* phase) and $H_1$ is rich in the solvent (*extract* phase). The line $G_1H_1$ is called a *tie line*. Also shown are two other tie lines $G_2H_2$ and $G_3H_3$. Since such a line connects two points representing two phases in equilibrium, it is called a *tie line* (see Section 7.1.2). There is an infinite number of such tie lines corresponding to an equilibrium diagram. Usually, these are not parallel. The length of a tie line decreases as we move away from the side $AB$, and eventually the last tie line reduces to a point $P$. The point $P$, that demarcates the raffinate and the extract sides of the equilibrium curve, is called the *plait point*. The type of equilibrium curve $RPS$ is called *binodal* because it has two arms $RP$ and $PS$, representing the raffinate and the extract sides.

The equilibrium curve also demarcates the miscible and immiscible regions on the graph. A liquid mixture having an 'overall composition' corresponding to a point $K$ below the equilibrium

curve *RPS* is a two-phase mixture. Such a mixture, when left undisturbed, separates into two phases at equilibrium. The compositions of these two phases are given by the terminals ($G_2$ and $H_2$) of the tie line passing through the point *K*; their amounts are given by the 'Lever Arm Rule' [see Eq. (7.4)].

The equilibrium curve is different for a Type II system; Figure 8.3(c) shows a representative equilibrium diagram. The raffinate and the extract arms of the equilibrium diagram do not meet. So there is *no plait point*. A few tie lines are shown in the figure.

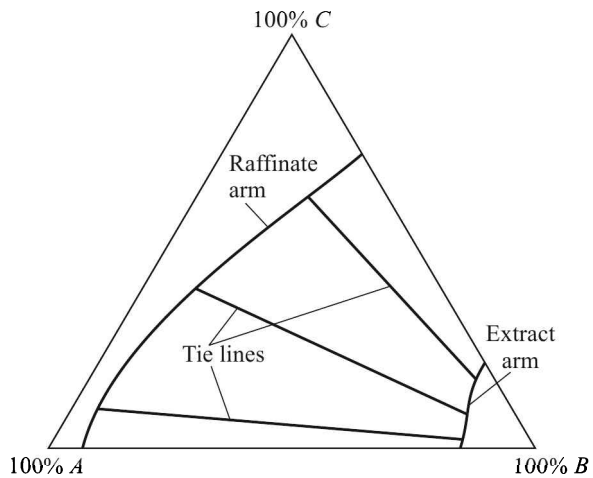


**Figure 8.3(c)** Liquid–liquid equilibrium for a type II system (*A–B* and *B–C* partially miscible).

### *The right-angled triangular diagram*

The rectangular coordinate system is used to prepare such a plot. Mass fractions of the solute in the two phases (the raffinate and the extract phases) at equilibrium are plotted against the corresponding mass fractions of the solvent [Figure 8.4(a)]. Concentrations of the extract phase (usually in mass fraction) are denoted by $y_A$, $y_B$, and $y_C$; those of the raffinate are $x_A$, $x_B$, and $x_C$. The raffinate arm *PQ* is obtained by plotting the set of points ($x_B$, $x_C$); the extract arm *PR* is

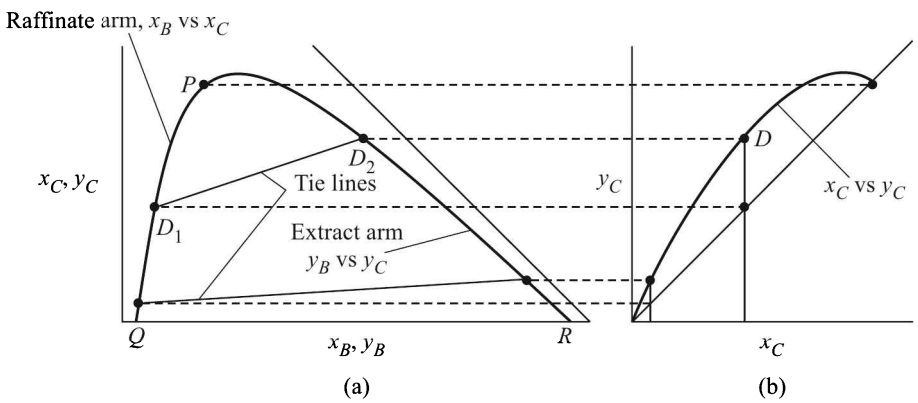


**Figure 8.4** (a) Equilibrium plot on the right triangular coordinate, and (b) the *x–y* plot.

obtained by plotting the set of points ($y_B$, $y_C$). The point $P$ is the plait point. The point $D_1$ on the raffinate arm and $D_2$ on the extract arm represent compositions at equilibrium. If they are joined, we get the tie line $D_1D_2$. The equilibrium diagram can also be drawn on the $x_C$–$y_C$ plane [Figure 8.4(b)]. The tie line $D_1D_2$ corresponds to the point $D$ on that plane.

### *Equilibrium plot on solvent-free basis*

In this presentation of LLE data, the mass ratios of the solute ($X$ and $Y$) in the two phases are plotted versus the corresponding mass ratios ($z$ or $Z$) of the solvent, all on 'solvent-free basis'. The quantities ($X$, $Y$, and $z$ or $Z$) are defined below.

$$X = \frac{x_C}{x_A + x_C}, \qquad z = \frac{x_B}{x_A + x_C} \qquad \text{for the raffinate}$$

$$Y = \frac{y_C}{y_A + y_C}, \qquad Z = \frac{y_B}{y_A + y_C} \qquad \text{for the extract}$$

Here $x$ is the mass fraction of a component in the raffinate, and $y$ is that in the extract. A typical equilibrium plot on "solvent-free basis" is shown in Figure 8.5. The $X$–$z$ and $Y$–$Z$ plots of LLE are similar to the enthalpy–concentration diagrams ($x$–$H_L$ and $y$–$H_V$) used in distillation calculations. Such plots ($X$–$z$ and $Y$–$Z$) are called *Maloney-Schubert diagram* or *Janecke diagram*.

**EXAMPLE 8.1** (*Janecke diagram*) Draw the Janecke diagram for a liquid–liquid system using the following equilibrium data. Show the tie lines on the diagram.

| Raffinate phase | | | Extract phase | | |
|---|---|---|---|---|---|
| $x_A$ | $x_B$ | $x_C$ | $y_A$ | $y_B$ | $y_C$ |
| 0.939 | 0.0601 | 0.0 | 0.06 | 0.94 | 0.0 |
| 0.7327 | 0.0654 | 0.0219 | 0.0520 | 0.918 | 0.03 |
| 0.4402 | 0.0732 | 0.4866 | 0.036 | 0.88 | 0.084 |
| 0.246 | 0.090 | 0.664 | 0.0185 | 0.851 | 0.1305 |
| 0.103 | 0.1047 | 0.7923 | 0.0069 | 0.8276 | 0.1655 |
| 0.0 | 0.1103 | 0.8897 | 0.0 | 0.8246 | 0.1754 |

### *Solution*

The $X$–$z$ and $Y$–$Z$ data are calculated from the above table.

$$X = \frac{x_C}{x_A + x_C} \quad 0.0 \quad 0.0234 \quad 0.525 \quad 0.730 \quad 0.885 \quad 1.0$$

$$z = \frac{x_B}{x_A + x_C} \quad 0.064 \quad 0.067 \quad 0.079 \quad 0.099 \quad 0.117 \quad 0.124$$

(*for the raffinate phase*)

$$Y = \frac{y_C}{y_A + y_C} \quad 0.0 \quad 0.366 \quad 0.70 \quad 0.876 \quad 0.960 \quad 1.0$$

$$Z = \frac{y_B}{y_A + y_C} \quad 15.67 \quad 11.2 \quad 7.33 \quad 5.71 \quad 4.8 \quad 4.7$$

(*for the extract phase*)

The $X$–$z$ and $Y$–$Z$ plots and the tie lines are shown in the Janecke diagram, Figure 8.5.

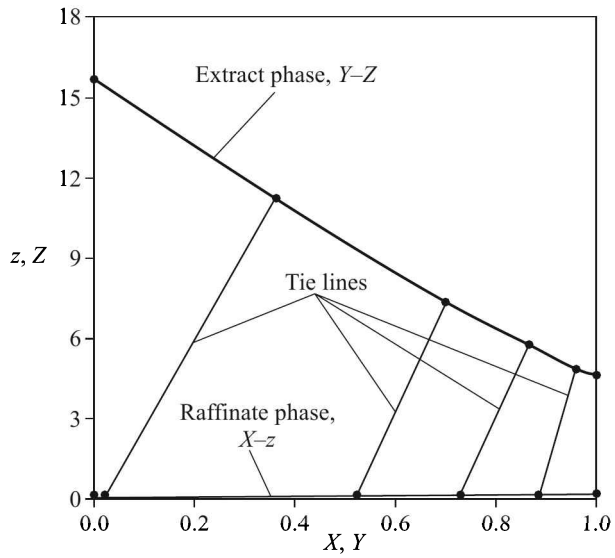


**Figure 8.5** Janecke plot of liquid–liquid equilibrium data. (A few tie lines are also shown.)

### 8.2.4 Effect of Temperature on Liquid–Liquid Equilibria

The effect of temperature on liquid–liquid equilibria shown in Figure 8.6 is pretty strong. Miscibility of a liquid pair generally increases with temperature. With increasing temperature, the miscibility of the *B*–*C* pair (the solvent and the solute) in a Type II system increases more quickly than that of the *A*–*B* pair. At the critical solution temperature of the *B*–*C* pair, a Type II system

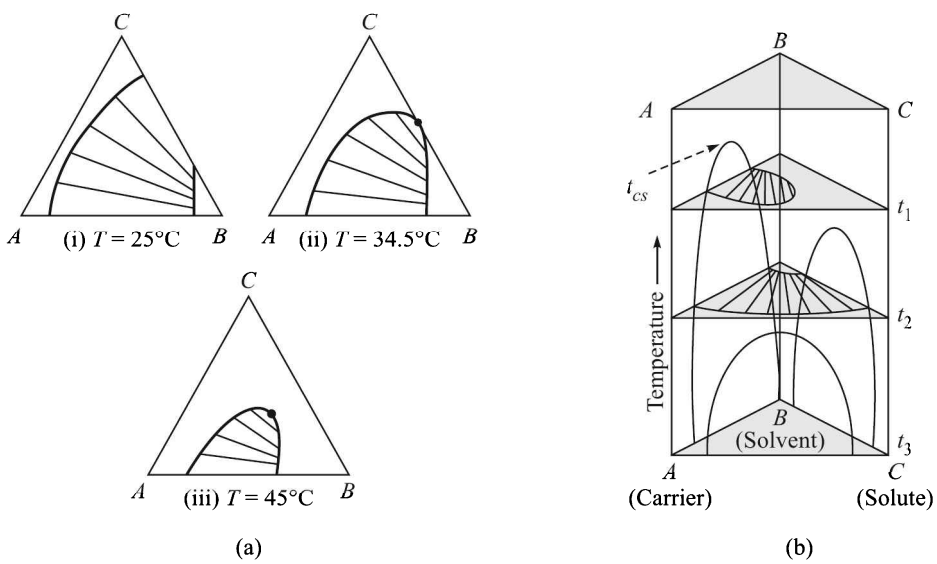


**Figure 8.6** Effect of temperature on liquid–liquid equilibria: (a) effect of temperature on solubility for the system *n*-hexane (*A*)–methyl-cyclohexane (*C*)–aniline (*B*), and (b) temperature effect illustrated by a three-dimensional sketch.

turns into a Type I system. This behaviour is displayed by the *n*-hexane(*A*)–methylcyclohexane(*C*)–aniline(*B*) system with increasing temperature [Figure 8.6(a)]. The two-phase region of the equilibrium diagram for a Type II system still contracts with increasing temperature and eventually vanishes at a temperature called its 'critical solution temperature', $t_{cs}$ [Figure 8.6(b)]. The critical solution temperature is the minimum temperature at which the components *A*, *B* and *C* of a Type I system becomes miscible in all proportions.

## 8.3 SOLVENT SELECTION

The right solvent is the key to successful separation by solvent extraction (Cusack et al., 1991; Cusack and Glatz, 1996). Solvents used are mostly organic. The important criteria for solvent selection are given below. The recent computational strategies of molecular interaction for solvent selection in diverse applications have been reviewed by Gani et al. (2006).

(a) *Selectivity:* 'Selectivity' or preferential uptake of the solute by the solvent over the carrier is the single most important factor in choosing a solvent for an extraction job. Selectivity or 'separation factor' is defined as

$$\beta_{C,A} = \frac{y_C/y_A}{x_C/x_A} \tag{8.1}$$

Selectivity of the solute varies with the compositions of the two phases in equilibrium and depends on temperature. It is unity at the plait point because the compositions of the two phases become the same. A theoretical approach to selection of a solvent for a particular type or class of solute was proposed by Robbins (see Cusack et al., 1991). The basis of the approach is the following simple equality of the activities of the solute in the two phases at equilibrium.

$$x_C(\gamma_C)_R = y_C(\gamma_C)_E \quad \Rightarrow \quad K = \frac{y_C}{x_C} = \frac{(\gamma_C)_R}{(\gamma_C)_E} \tag{8.2}$$

where $(\gamma_C)_E$ and $(\gamma_C)_R$ are the activity coefficients of the solute in the extract and the raffinate phases respectively, and *K* is the 'distribution coefficient' or 'partition coefficient' of the solute for the carrier-solvent pair. It appears from Eq. (8.2) that if the activity coefficient of the solute is small in the extract phase, the distribution coefficient is large, and the solvent has a better selectivity for the particular solute. Analyzing the equilibrium data of a large number of systems (more than 900), Robbins (1980) prepared a qualitative summary (Table 8.2) of the solvent–solute interaction in terms of its effect on the activity coefficient. The functional groups of solute are numbered and named in the left two columns of Table 8.2. The numbers given in the top row also represent solvents having the corresponding functional groups. The rest of the table is a matrix having +, – and 0 as the elements. A '+' sign at a location indicates that compounds belonging to the particular group of solvents in the column tend to raise the activity coefficient of the solute belonging to the group in the row. For example, a solute having a phenolic group (group 1 of the solute class) exhibits a high activity in a solvent (like a paraffin) belonging to group 11 or 12 (the 1,11 and 1,12 positions of table have a '+' sign). But in a solvent of group 5, the activity coefficient of the same solute is lower (the 1,5 position has a '–' sign). Thus for extraction of phenol, a ketone is a good solvent because the activity of phenol in a ketone is low and the distribution coefficient is predicted to be large. Table 8.2 classifies solvents primarily on the basis of hydrogen bonding and electron donor–acceptor interaction with solutes.

**Table 8.2** Solute–solvent group interactions—preliminary guide to solvent selection#

| Solute class | Group | Solvent class 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. | Phenol | 0 | 0 | – | 0 | 0 | – | – | – | – | – | – | – |
| 2. | Acid, thiol | 0 | 0 | – | 0 | – | – | 0 | 0 | 0 | 0 | + | + |
| 3. | Alcohol, water | – | – | 0 | + | + | 0 | – | – | + | + | + | + |
| 4. | Active-H on multihalo paraffin | 0 | 0 | + | 0 | – | – | – | – | – | – | 0 | + |
| | H-acceptor | | | | | | | | | | | | |
| 5. | Ketone, amide with no H on N, sulphone, phosphine oxide | – | – | + | – | 0 | + | + | + | + | + | + | + |
| 6. | Tertiary amine | – | – | 0 | – | + | 0 | + | + | 0 | + | 0 | 0 |
| 7. | Secondary amine | – | 0 | – | – | + | + | 0 | 0 | 0 | 0 | 0 | + |
| 8. | Primary amine, ammonia, amide with 2H on N | – | 0 | – | – | + | + | 0 | 0 | + | + | + | + |
| 9. | Ether, oxide, sulphoxide | – | 0 | + | – | + | 0 | 0 | + | 0 | + | 0 | + |
| 10. | Ester, aldehyde, carbonate phosphate, nitrate, nitrite nitrile, intramolecular bonding (for example, *o*-nitro phenol) | – | 0 | + | – | + | + | 0 | + | + | 0 | + | + |
| 11. | Aromatic, olefin, halogen aromatic multi-halo-paraffin without active H, mono-halo-paraffin | + | + | + | 0 | + | 0 | 0 | + | 0 | + | 0 | 0 |
| | Non-H-bonding | | | | | | | | | | | | |
| 12. | Paraffin, carbon disulphide | + | + | + | + | + | 0 | + | + | + | + | 0 | 0 |

*Key:* Deviation from normality for mixtures (+) = positive; (–) = negative; (0) = neutral

#*Source:* Robbins, L.A., *Chem. Eng. Prog.*, 76, October 1980, pp. 58–61.

(b) *Carrier-solvent immiscibility:* If the mutual solubility of the carrier and the solvent is low, they can be satisfactorily separated in the settler. Only a small quantity of the solvent will be retained in the raffinate and a small quantity of the carrier will remain dissolved in the solvent. So the cost of further separation of the phases by distillation or otherwise will be small.

(c) *Interfacial tension:* Interfacial tension between the carrier and the solvent has two opposing effects. A low interfacial tension favours fine dispersion of one liquid in the other, thereby increasing the specific interfacial area of mass transfer. The power consumption for dispersing one liquid into the other is also lower. On the other hand, coalescence of droplets leading to the separation of the raffinate and the extract phases does not occur readily if the interfacial tension is low. It is desirable to strike a balance between these opposing effects while selecting a solvent.

(d) *Density:* The driving force for phase separation in a settler is the difference in the densities of the two phases. The density of the lighter phase should generally be at least 5% less than that of the heavier phase in order to ensure smooth phase separation.

(e) *Viscosity:* The solvent viscosity should be low. This reduces power consumption for mixing the phases.

(f) *Other factors:* The solvent should preferably be less expensive, non-toxic, and non-corrosive. It should have a low flamability and volatility. Another very important phenomenon is accumulation of impurities in the solvent. If the solvent is of low volatility, it leaves the solvent clean-up tower at the bottom (Figure 8.1) and is recycled. There is a strong probability of accumulation of impurities and surface active agents. This causes lowering of interfacial tension and surface resistance to mass transfer between the two phases. The performance of an extractor may be greatly reduced by such phenomena. It has been estimated (see *Chem. Engg. Progress*, June 2002, p. 74) that the presence of impurities can reduce the capacity of a column by 20% or more and its efficiency by as much as 60%.

Hardly any solvent will satisfactorily meet all the above criteria. Choosing a solvent is again a matter of striking a balance between the properties and cost of the solvent on the one hand, and the equipment and operating costs and ease of operation on the other.

## 8.4 DESIGN CALCULATIONS FOR STAGE-WISE EXTRACTION

While dealing with the problem of separation of a liquid mixture, a decision has to be taken first about the separation process (such as distillation, extraction, crystallization, etc.) to be adopted. If liquid–liquid extraction appears to be the choice, the next job is to select the solvent to be used. The design of extractors is far less standardized than that of absorption or distillation columns. Pilot plant tests may have to be done if reliable design data collected from operating units handling a similar mixture are not available (Cusack and Glatz, 1998).

Like other mass transfer operations, solvent extraction can be carried out in a staged or in a continuous contact equipment (see Section 8.6). The important stage-wise pieces of contact equipment are (i) mixer-settler, (ii) sieve tray column, and (iii) Schiebel column. Continuous contact takes place in an equipment like (i) the rotating disk contactor (RDC), (ii) Graesser raining bucket contactor, (iii) Podbielniak extractor, (iv) the packed column, and (v) the spray column, etc. In this section, we will discuss the methods of determination of the number of ideal stages required for solvent extraction in a stage-wise contact equipment. The methods to be described here are based on graphical construction of ideal stages for partially miscible ternary systems. Determination of the number of ideal stages for extraction systems in which the solvent and the carrier (the *A–B* pair) are practically immiscible can be done by following the procedures already described in Chapter 4.

### 8.4.1 Single-Stage Operation

A single-stage equilibrium contact device is shown in Figure 8.7(a). The feed enters the mixer at a rate $F$ having a solute concentration $x_F$ (the solute concentration in the raffinate is denoted by $x$ and that in the extract by $y$, both in mass%). The solvent is supplied at a rate of $S$; it has a solute concentration $y_S$[⊕]. The liquid–liquid dispersion attains equilibrium and leaves the mixer at a rate $M$ with an overall solute concentration $x_M$. The raffinate (flow rate = $R$, solute

[⊕] The solute is separated from the extract (solvent) phase (as shown in Figure 8.1) and the solvent is recycled. So the feed solvent is likely to have a little solute left in it.

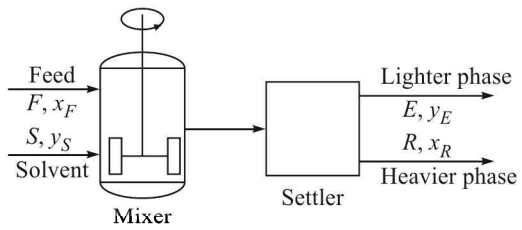


**Figure 8.7(a)** Schematic of a single-stage extraction unit.

concentration = $x_R$) and the extract (flow rate = $E$, solute concentration = $y_E$) phases leave the settler after phase separation. The following material balance equations may be written.

Total material balance: $$F + S = M = E + R \tag{8.3}$$

Solute balance: $$Fx_F + Sy_S = Mx_M = Ey_E + Rx_R \tag{8.4}$$

$$\Rightarrow \quad x_M = \frac{Fx_F + Sy_S}{F + S} \tag{8.5}$$

Combining the above two equations, we may write

$$Fx_F + Sy_S = (F + S)x_M$$

$$\Rightarrow \quad \frac{F}{S} = \frac{x_M - y_S}{x_F - x_M} \tag{8.6}$$

The ternary equilibrium data are plotted on a suitable diagram. Let us select the equilateral triangular diagram [Figure 8.7(b)]. The compositions of the feed and the solvent streams are located at points $F$ and $S$ on the diagram†. By virtue of the Lever Arm Rule (Section 7.4), we may conclude that the mixture of mass $M$ and solute concentration $x_M$ can be represented by a point $M$ on the line joining $F$ and $S$. From Eqs. (8.3) and (8.4), we may also write

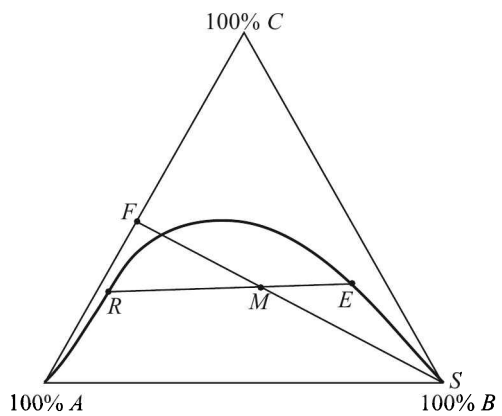


**Figure 8.7(b)** Single-stage liquid extraction calculation using the equilibrium plot on the right triangular coordinate.

$$Rx_R + Ey_E = (R + E)x_M \quad \Rightarrow \quad \frac{E}{R} = \frac{x_M - x_R}{y_E - x_M} \tag{8.7}$$

The above equation indicates that the point $M$ also lies on the line joining the points $E$ and $R$. Since the mixer acts as a 'theoretical' or 'equilibrium stage', the phases $E$ and $R$ are at equilibrium and must lie on the terminals of the tie line through the point $M$ on the equilateral triangular diagram. Thus the solute concentrations in the extract and the raffinate phases ($y_E$ and $x_R$) can

† By convention the symbols $F$ and $S$ represent the flow rates and locations of the two points representing their compositions. Similar convention has been used in Section 7.2.

be obtained by drawing the tie line through the point $M$. The steps of the graphical procedure are given below.

(a) Draw the ternary equilibrium curve on the equilateral triangular diagram (or any other suitable diagram). Locate the points $F$ and $S$.
(b) Calculate $x_M$ from the material balance equation [Eq. (8.5)] and locate the point $M$ on the line $FS$.
(c) Draw the tie line $ER$ through the point $M$. The terminals of the tie line give the extract and raffinate concentrations, $y_E$ and $x_R$. The amount of the extract and raffinate phases ($E$ and $R$) and the percentage of the solute separated from the feed can now be calculated from Eqs. (8.3) and (8.4) or Eq. (8.7).

If a required percentage separation of the solute is specified, the necessary solvent rate can be calculated. However, this will involve a trial-and-error procedure. This is illustrated in Example 8.2 using the right triangular coordinate.

**EXAMPLE 8.2** (*Single-stage extraction*) One thousand kilograms of an aqueous solution containing 50% acetone is contacted with 800 kg of chlorobenzene containing 0.5 mass% acetone in a mixer-settler unit, followed by separation of the extract and the raffinate phases. (a) Determine the composition of the extract and the raffinate phases and the fraction of acetone extracted. (b) Also calculate the amount of solvent required if 90% of the acetone is to be removed. Equilibrium and tie line data are given below.

| Aqueous phase (*Raffinate*) | | | Organic phase (*Extract*) | | |
|---|---|---|---|---|---|
| Water | Chlorobenzene | Acetone | Water | Chlorobenzene | Acetone |
| $x_A$ | $x_B$ | $x_C$ | $y_A$ | $y_B$ | $y_C$ |
| 0.9989 | 0.0011 | 0.0 | 0.0018 | 0.9982 | 0.0 |
| 0.8979 | 0.0021 | 0.1 | 0.0049 | 0.8872 | 0.1079 |
| 0.7969 | 0.0031 | 0.2 | 0.0079 | 0.7698 | 0.2223 |
| 0.6942 | 0.0058 | 0.3 | 0.0172 | 0.608 | 0.3748 |
| 0.5864 | 0.0136 | 0.4 | 0.0305 | 0.4751 | 0.4944 |
| 0.4628 | 0.0372 | 0.5 | 0.0724 | 0.3357 | 0.5919 |
| 0.2741 | 0.1259 | 0.6 | 0.2285 | 0.1508 | 0.6107 |
| 0.2566 | 0.1376 | 0.6058 | 0.2566 | 0.1376 | 0.6058 |

### *Solution*

*Given:* mass of the feed, $F$ = 1000 kg; mass fraction acetone ($C$) in the feed, $x_{C,F}$ = 0.5; and the mass fraction of chlorobenzene in the feed, $x_{B,F}$ = 0.

**(a)** Solvent: mass, $S$ = 800 kg; $y_{C,S}$ = 0.005; $y_{B,S}$ = 0.995
From Eqs. (8.3) and (8.5),

$$M = F + S = 1800; \qquad x_{M,C} = \frac{F\,x_{C,F} + S\,y_{C,S}}{M} = \frac{(1000)(0.5) + (800)(0.005)}{1800} = 0.28$$

The equilibrium data are plotted using the right triangular coordinates [Figure 8.8(a)]. The points $F$ and $S$ are plotted on the diagram; the point $M$ is located on the line $FS$ (for $x_{M,C}$ = 0.28). The tie line $RE$ through $M$ is determined by trial taking help of the $x_C$–$y_C$ curve [Figure 8.8(b)] drawn

alongside on the same scale. It is to be noted that a point on the $x_C$–$y_C$ curve gives the equilibrium concentrations of the solute ($C$) in the two phases. The point $P$ on the $x_C$–$y_C$ diagram corresponds to the tie line $RE$.

From Figure 8.8(a) or 8.8(b), the solute concentrations in the raffinate ($R$) and the extract ($E$) phases are:

$$x_{C,R} = 0.236 \quad \text{and} \quad y_{C,E} = 0.302$$

From Eqs. (8.3) and (8.7),

$$E + R = 1800 \quad \text{and} \quad \frac{E}{R} = \frac{0.28 - 0.236}{0.302 - 0.28} \quad \Rightarrow \quad E = \boxed{1200 \text{ kg}} \quad R = \boxed{600 \text{ kg}}$$

Mass of acetone leaving with the raffinate = $(R)(x_{C,R})$ = (1200)(0.302) = $\boxed{362.4 \text{ kg}}$

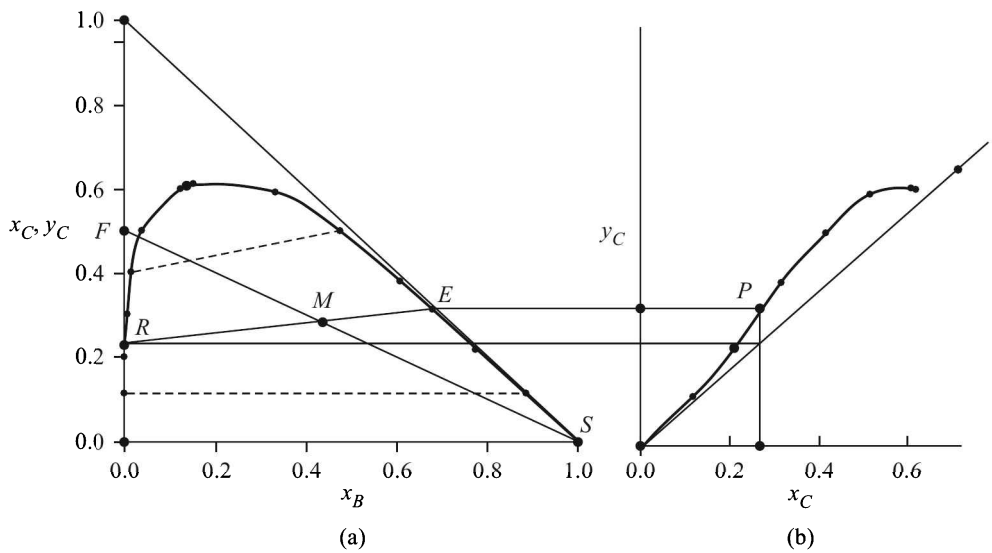


**Figure 8.8** Single-stage extraction calculation illustrated: (a) Right triangular coordinate, and (b) $x$–$y$ diagram.

Mass of acetone in the feed = (1000)(0.5) = 500 kg

Acetone entering with the solvent = (800)(0.005) = 4 kg

Net amount of acetone transported to the extract stream = 362.4 – 4 = 358.4 kg

$$\text{Fraction of acetone removed by extraction} = \frac{358.4}{500} \quad \text{i.e.} \quad \boxed{71.7\%}$$

**(b)** If 90% of the acetone is removed,

acetone remaining in the aqueous phase = (500)(0.1) = 50 kg

It is to be noted that at this low concentration of the solute, the mutual solubility of water and chlorobenzene is negligible. Therefore, the raffinate may be assumed to be virtually free from chlorobenzene. Then the concentration of acetone in the raffinate is 50/550 = 0.0909.

At $x_{C,R}$ = 0.0909, we have $y_{C,E}$ = 0.105 (from the equilibrium curve).

Also, the mass of the raffinate, $R$ = 550 kg

If $S$ is the mass of the solvent required, we can write the following material balance for the solute.

$$(1000)(0.5) + (S)(0.005) = (S + 450)(0.105) + (550)(0.0909) \Rightarrow S = \boxed{4027 \text{ kg}}$$

### 8.4.2 Determination of the Tie Line Through a Point on the Equilibrium Line

Only a few experimental tie line data for a ternary system are generally available. There are two 'interpolation' techniques of drawing a tie line through any point on the equilibrium curve. Both the techniques use the available tie line data and a 'conjugate line' (Sherwood and Pigford, 1952; Seader and Henley, 1998). One technique is illustrated in Figure 8.9. A horizontal line is drawn through the end $G$ of the tie line $GH$. A vertical line is drawn through the point $H$. The two lines meet at $K$. The locus of the point $K$ is called the 'conjugate line'. Now let us use this conjugate line to draw a tie line through any point $G_1$ on the equilibrium curve. A horizontal line is drawn from $G_1$ that meets the conjugate line at $K_1$. A vertical line through $K_1$ meets the equilibrium curve at $H_1$. Then $G_1H_1$ is the tie line through the point $G_1$.

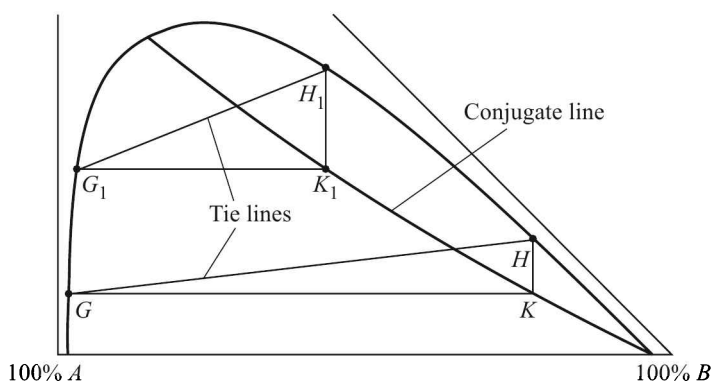


**Figure 8.9** Construction of a conjugate line using the tie-line data.

### 8.4.3 Multistage Cross-current Extraction

A multistage cross-current cascade is schematically shown in Figure 8.10. The solvent is fed to the stages separately and the amounts need not be equal. The feed is supplied to the stage at one

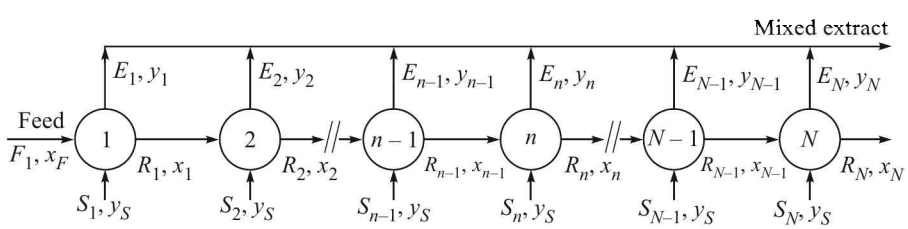


**Figure 8.10** Schematic of a cross-current extraction cascade.

end of the cascade (let us call it stage 1). It moves from one stage to the next in succession with gradually decreasing solute concentration. If the solvent and the carrier liquid have negligible miscibility, extraction calculations can be done following the procedure described in Section 4.6.4. Here we shall deal with ternary systems in which the carrier and the solute have limited solubility.

The graphical method of calculation is convenient. Any of the three types of equilibrium diagrams (drawn using equilateral triangular, right triangular or 'solvent-free' coordinates) may be used. But the calculation procedures are similar.

Let there be $N$ number of stages in the cross-current extraction cascade. The flow rates and concentrations of all the input and output streams are shown in Figure 8.10. We may write the following mass balance equations for the $n$th stage.

Total mass balance : $$R_{n-1} + S_n = M_n = E_n + R_n \tag{8.8}$$

Mass balance of the solute $C$ : $$R_{n-1}x_{n-1} + S_n y_S = M_n x_{M,n} = E_n y_n + R_n x_n \tag{8.9}$$

The principle of multistage extraction calculation is similar to that of a single-stage extraction unit. From Eq. (8.8), we see that if the flow rates and concentrations of the streams $R_{n-1}$ and $S_n$ are known, the point $M_n$ can be located on the line joining $R_{n-1}$ and $S_n$. Again, if the tie line through the point $M_n$ is drawn, its terminals give the points $R_n$ and $E_n$. A step-by-step procedure is given below.

(a) Stage 1 receives the fresh feed which does not generally contain any solvent. Putting $n = 1$, in Eq. (8.8), we get the material balance equation for stage 1.

$$F + S_1 = M_1 = E_1 + R_1 \tag{8.10}$$

The point $F$ is located on the $x$–$y$ plane. The point $S_1$ is also located. The point $M_1$ is located on the line joining $F$ and $S_1$ by using the Lever Arm Rule or by calculating the value of $x_{M1}$ from Eq. (8.9). The tie line through the point $M_1$ is drawn and the points $E_1$ and $R_1$ are located at its terminals.

(b) For stage 2, the solvent rate is $S_2$ and the rate of input of feed is $R_1$(which is nothing but the 'first-stage raffinate'). The point $S_2$ is located on the graph; the other point $R_1$ is known. These two points are joined and the mixture of the phases $M_2$ is located on the line joining these points $R_1$ and $S_2$. Again the terminals of the tie line through $M_2$ give the points $E_2$ and $R_2$.

(c) The procedure is repeated for the remaining stages. In the above analysis, we have tacitly assumed that the rates of solvent input to the stages and its concentrations are known. The flow rates and concentrations of the 'raffinates' and 'extracts' from all the stages can be found out by following the above steps. In practical problems, the solute concentrations, if any, in the feed solvent streams to the different stages remain the same. The procedure is illustrated in Example 8.3.

There may be other versions of the design problem. For example: the feed rate and composition, the solvent composition and the maximum allowable concentration of the solute in the final raffinate are given; the same amount of solvent is fed to the stages; it is required to determine the solvent rate if the extraction is to be done in a specified number of stages. This and other versions of the design problem may be solved using the principles discussed above. But trial-and-error calculations may sometimes be necessary depending upon the nature of the problem.

**EXAMPLE 8.3** (*Cross-current extraction calculation*) One thousand kilograms of an aqueous solution containing 35 mass% trimethyl amine (TMA) and 65% water is to be extracted using benzene as the solvent. A three-stage cross-current extraction scheme is suggested. The amounts of solvent (98% benzene, 2% TMA) to be used in successive stages are 815 kg, 950 kg and 2625 kg. Determine the fraction of the solute removed if the stages are ideal. The compositions of the raffinate and the extract (two phases) as well as the tie-line data are given below (water: *A*; benzene: *B*; TMA: *C*).

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Water-rich phase: | $x_B$ | 0.004 | 0.006 | 0.01 | 0.02 | 0.03 | 0.036 | 0.07 | 0.13 |
| | $x_C$ | 0.05 | 0.10 | 0.15 | 0.20 | 0.35 | 0.30 | 0.35 | 0.40 |
| Benzene-rich phase: | $y_B$ | 0.95 | 0.90 | 0.84 | 0.78 | 9.71 | 0.63 | 0.50 | 0.26 |
| | $y_C$ | 0.05 | 0.10 | 0.15 | 0.20 | 0.25 | 0.30 | 0.35 | 0.40 |
| Tie-line data: | $x_C$ | 0.04 | 0.083 | 0.13 | 0.215 | 0.395 | | | |
| | $y_C$ | 0.035 | 0.068 | 0.09 | 0.145 | 0.31 | | | |

### *Solution*

The problem is solved using the right triangular coordinate.

Feed, $F$ = 1000 kg. Feed composition: 35% TMA ($x_{C,F}$ = 0.35); 65% water ($x_{A,F}$ = 0.65).

The equilibrium data are plotted and the tie lines are drawn (dotted lines) in Figure 8.11.

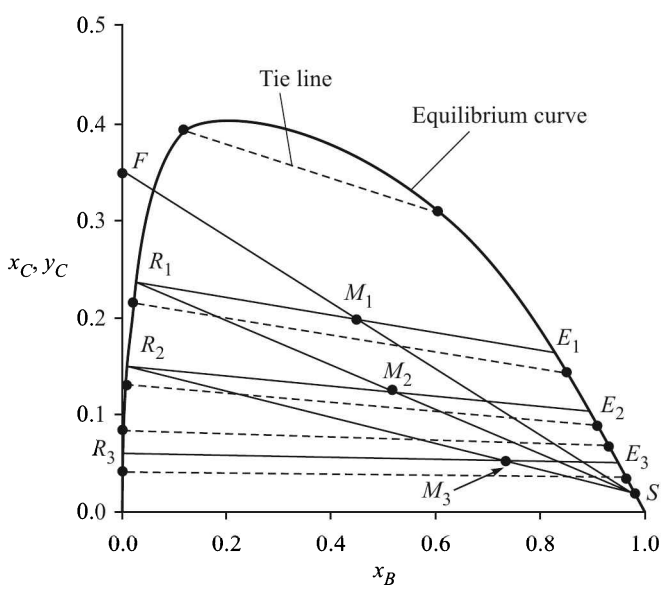


**Figure 8.11** Graphical solution for a three-stage cross-current extraction cascade.

**Stage 1:** The solvent rate, $S_1$ = 815 kg; $y_{B,S}$ = 0.98, $y_{C,S}$ = 0.02 (2% TMA, rest benzene).

$$M_1 = 1000 + 815 = 1815 \text{ kg}; \quad x_{C,M1} = [(1000)(0.35) + (815)(0.02)]/1815 = 0.202$$

The points $F$ ($x_{B,F}$ = 0, $x_{C,F}$ = 0.35), $S$ ($y_{B,S}$ = 0.98, $y_{C,S}$ = 0.02) are located on the diagram. The point $M_1$ is located on the line $FS$ ($x_{C,M1}$ = 0.202). The tie line $R_1E_1$ is drawn; the extremities of the tie line give the compositions of the raffinate and extract phases leaving stage 1 in equilibrium.

From the figure, $x_{C,R1} = 0.24$; $y_{C,E1} = 0.166$

Total material balance: Eq. (8.3), $M_1 = R_1 + E_1 = 1815$ (i)

Solute balance: $(M_1)(x_{C,M1}) = (R_1)(x_{C,R1}) + (E_1)(y_{C,R1})$

$\Rightarrow$ $(1815)(0.202) = (R_1)(0.24) + (E_1)(0.166)$ (ii)

Solving the material balance equations (i) and (ii), $R_1 = \boxed{882 \text{ kg}}$; $E_1 = \boxed{933 \text{ kg}}$

**Stage 2:** The raffinate from stage 1 ($R_1$) is the feed to stage 2. Solvent added, $S_2 = 950$ kg; $y_{C,S2} = 0.02$.

$M_2 = R_1 + S_2 = 882 + 950 = 1832$ kg; solute balance: $M_2 x_{C,M2} = (R_1)(x_{C,R1}) + (S_2)(y_{C,S2})$

Putting the values of different quantities, $x_{C,M2} = 0.126$.

The composition of the solvent to the second stage being the same as that to the first, it is represented by the same point $S$. The line $R_1S$ is joined and the point $M_2$ ($x_{C,M2} = 0.126$) is located on this line as shown in Figure 8.11. The tie line $R_2E_2$ is drawn through the point $M_2$.

The compositions of the raffinate and the extract phases from stage 2 are obtained from the figure as: $x_{C,R2} = 0.15$ and $y_{C,E2} = 0.105$

Material balance equations similar to Eqs. (i) and (ii) are written. Solving these equations, we get

$$R_2 = \boxed{854.3 \text{ kg}} \qquad E_2 = \boxed{977.7 \text{ kg}}$$

**Stage 3:** Similarly, the third stage receives $R_2 = 854.3$ kg 'feed' and $S_3 = 2625$ kg of the solvent. Proceeding in the same way we can locate the point $M_3$ ($x_{C,M3} = 0.052$) on the line $R_2S$ and draw the tie line through $M_3$ to meet the equilibrium curve at the points $R_3$ ($x_{C,R3} = 0.0596$) and $E_3$ ($y_{C,E3} = 0.0506$). The amounts of the phases can be obtained from material balance as before:

$$R_3 = \boxed{540.8 \text{ kg}} \qquad \text{and} \qquad E_3 = \boxed{2938.5 \text{ kg}}$$

Total amount of TMA leaving the stages

$$= (E_1)(y_{C,E1}) + (E_2)(y_{C,E2}) + (E_3)(y_{C,E3})$$
$$= (933)(0.166) + (977.7)(0.105) + (2938.5)(0.0506) = 406.2 \text{ kg}$$

Total amount of TMA entering the three stages with the solvent

$$= (S_1 + S_2 + S_3)(y_{C,S}) = (815 + 950 + 2625)(0.02) = 87.8 \text{ kg}$$

Net amount of TMA removed = 406.2 – 87.8 = 318.4 kg

Total amount of solute in the feed = (1000)(0.35) = 350 kg

Fraction of TMA removed = (318.4/350)(100) = $\boxed{91\%}$

### 8.4.4 Countercurrent Extraction—Determination of the Number of Ideal Stages

A countercurrent extraction cascade, along with the flow rates and concentrations of the input and output streams at the stages, is shown in Figure 8.12. The concentrations and flow rates

of the feed and the solvent are usually given. If the solvent rate is not specified, it can be taken as a suitable multiple of the 'minimum solvent rate'. We have to determine the number of ideal contact stages required if the maximum permissible concentration of the solute in the raffinate is given. We may write the following material balance equations.

Total material balance over the entire unit (over envelope I): $F + S = E_1 + R_N = M$ (8.11)

Solute balance over envelope I: $Fx_F + Sy_S = E_1y_1 + R_Nx_N = Mx_M$ (8.12)

Total material balance over envelope II: $R_{n-1} + S = E_n + R_N$ (8.13)

Solute balance over envelope II: $R_{n-1}x_{n-1} + Sy_S = E_ny_n + R_Nx_N$ (8.14)

The equilibrium curve is drawn using a suitable coordinate system. From the known compositions of the feed, the solvent and the raffinate, the points $F$, $S$ and $R_N$, can be located on the diagram. The point $M$ representing the mixture of $F$ and $S$ is also located on the line $FS$. Equation (8.11) says that the point $M$ lies also on the line joining the points $R_N$ and $E_1$ (the point $E_1$ represents the extract phase leaving stage 1; see Figure 8.12). Therefore, the line $R_NM$ when extended meets the equilibrium curve at the point $E_1$.

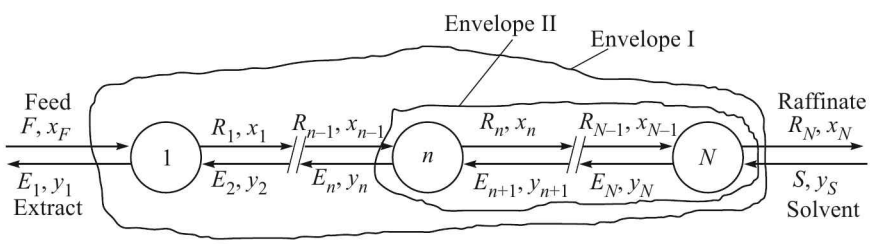


**Figure 8.12** Schematic of a countercurrent extraction cascade.

Now we rewrite Eq. (8.11) in the following form:

$$E_1 - F = S - R_N = \Delta \tag{8.15}$$

Also from Eqs. (8.13) and (8.15),

$$E_n - R_{n-1} = S - R_n = \Delta \tag{8.16}$$

Here $\Delta$ may be considered to be a fictitious stream obtained by 'removing' a mass $F$ of the feed from the first stage extract $E_1$, or by 'removing' a mass $R_N$ of the 'last stage raffinate' from the solvent $S$. It represents a 'difference point' (similar to that identified in Section 7.8) lying on the line $FE_1$ extended and also on $R_NS$ extended. In other words, it is the point of intersection of $FE_1$ and $R_NS$. Also, it follows from Eq. (8.16) that if the line $R_{n-1}E_n$ is extended, it has to pass through the difference point $\Delta$. These results can be directly used to determine the number of ideal contact stages required to achieve the given degree of separation.

Since the stages are ideal, the streams $E_n$ and $R_n$ leaving the $n$th stage are at equilibrium. Thus the tie line through the point $E_1$ meets the equilibrium curve at $R_1$. Equation (8.16) says that for $n = 2$, the line joining $R_1$ and $\Delta$ meets the equilibrium curve at $E_2$. The tie line through $E_2$ meets the equilibrium curve at the point $R_2$. The procedure is repeated till the point $R_N$ representing the raffinate is reached. The number of tie lines drawn in the process to reach the point $R_N$ is the number of ideal stages. The procedure is illustrated in Example 8.4.

**EXAMPLE 8.4** (*Multistage countercurrent extraction*) It is planned to extract diphenyl hexane (DPH) from a solution in docosane (*A*) using 'pure' furfural (*B*) as the solvent. The feed enters the extractor cascade at a rate of 2000 kg/h with 45% DPH (*C*) that has to be reduced to 4% in the final raffinate. The solvent rate is 2500 kg/h. Determine the number of theoretical stages required. Extraction is to be carried out at 45°C. Several compositions on the extract and the raffinate arms and the tie-line data in mass% of the components at 45°C are given below.

*Equilibrium data*

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *A:* | 96.0 | 84.0 | 67.0 | 52.5 | 32.6 | 21.3 | 13.2 | 7.7 | 4.4 | 2.6 | 1.5 | 1.0 | 0.7 |
| *B:* | 4.0 | 5.0 | 7.0 | 10.0 | 20.0 | 30.0 | 40.0 | 50.0 | 60.0 | 70.0 | 80.0 | 90.0 | 99.3 |
| *C:* | 0 | 11.0 | 26.0 | 37.5 | 47.4 | 48.7 | 46.8 | 42.3 | 35.6 | 27.4 | 18.5 | 9.0 | 0.0 |

*Tie-line data*

| Raffinate (Docosane) phase, mass% | | | Extract (Furfural) phase, mass% | | |
|---|---|---|---|---|---|
| *A* | *B* | *C* | *A* | *B* | *C* |
| 85.2 | 4.8 | 10.0 | 1.1 | 89.1 | 9.8 |
| 69.0 | 6.5 | 24.5 | 2.2 | 73.6 | 24.2 |
| 43.9 | 13.3 | 42.6 | 6.8 | 52.3 | 40.9 |

***Solution***

*Given:* $F = 2000$; $x_{C,F} = 0.45$; $x_{C,N} = 0.04$; $S = 2500$; $x_{C,S} = 0.0$

By Eq. (8.5),
$$x_{C,M} = \frac{F x_{C,F} + S y_{C,S}}{F + S} = \frac{(0.45)(2000) + (0)(2500)}{2000 + 2500} = 0.2$$

The raffinate (i.e. the stream $R_N$) has 4% DPH in it. So mass fraction of DPH in it is $x_C = 0.04$.

The equilibrium data are plotted using the right triangular coordinate in Figure 8.13(a); the tie lines are shown by light dotted lines. The points $F(0, 0.45)$, $S(1.0, 0)$, $M$(on the line $FS$ at $x_{C,M} = 0.2$), $R_N$ (at $x_C = 0.04$ on the equilibrium curve) are located. $R_N$ and $M$ are joined and the line is extended to meet the equilibrium curve at $E_1$. The lines $FE_1$ and $R_NS$, when extended, meet at the 'difference point', $\Delta$.

The stages being ideal, the tie line through $E_1$ meets the equilibrium curve at the point $R_1$, the raffinate phase leaving the first stage. Also, putting $n = 2$, it is evident from Eq. (8.16) that $E_2$ lies on the point of intersection of the line $R_1\Delta$ and the equilibrium curve. In fact, $R_1E_2$ is the 'operating line' for the second stage of the cascade. The tie line through the point $E_2$ meets the equilibrium curve at $R_2$. Proceeding in the same way, we can draw successive stages till the point $R_N$ is reached.

From Figure 8.13(a), the number of ideal stages required for the separation is 4.

An alternative technique can be more conveniently used for graphical construction of stages. A line emanating from the difference point, $\Delta$, intersects the extract arm of the LLE curve at a point $E_{n+1}$ [corresponding solute concentration is $y_C = (y_C)_{n+1}$] and the raffinate arm at a point $R_n$ [where $x_C = (x_C)_n$]. A few such lines (drawn as broken lines) are shown in Figure 8.13(b). The 'locus' of the point $[(x_C)_n, (y_C)_{n+1}]$ is the operating line on the $x_C$–$y_C$ plane [see

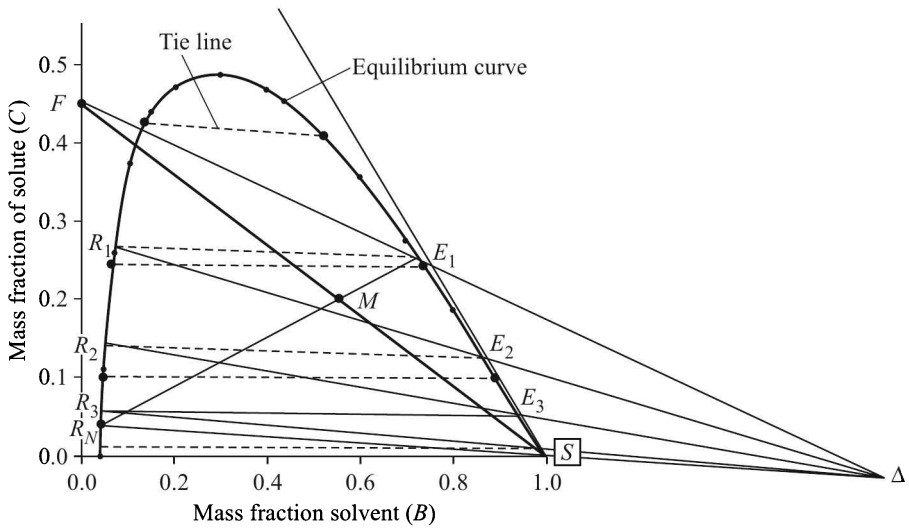


**Figure 8.13(a)** Graphical construction of ideal stages for countercurrent extraction using the right-angled triangular coordinate system.

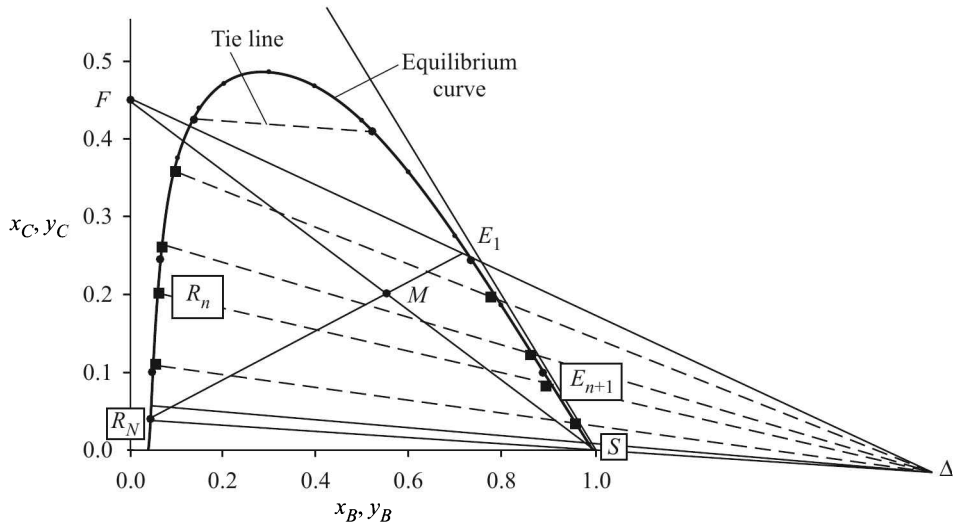


**Figure 8.13 (b)** Graphical construction for obtaining points on the operating line.

Figure 8.13(c)]. The set of ($x_C$, $y_C$) values derived from Figure 8.13(b) and used to draw the operating line on Figure 8.13(c) is given below.

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $x_C$ | 0.04 | 0.1 | 0.14 | 0.17 | 0.22 | 0.26 | 0.30 | 0.33 | 0.38 | 0.415 |
| $y_C$ | 0.03 | 0.051 | 0.018 | 0.09 | 0.125 | 0.15 | 0.178 | 0.20 | 0.252 | 0.280 |

The equilibrium line is drawn in Figure 8.13(c) using the given 'tie line' data. The operating line is also drawn by plotting the above $x_C$–$y_C$ data. Stages are now constructed by drawing stages

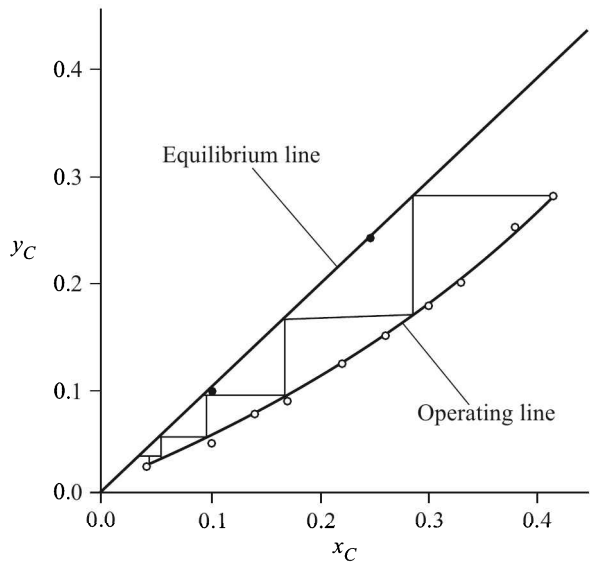


**Figure 8.13(c)** Construction of stages on the $x_C$–$y_C$ plane.

between the equilibrium line and the operating line. The number of ideal stages is 4.8 [compared to 4 obtained from Figure 8.13(a); the difference probably occurs because of the error in drawing the tie lines].

This technique of construction is simpler and gives a more accurate estimate of the number of stages and is recommended for use.

### 8.4.5 Stage-wise Extraction with Reflux

The solute concentration in the extract leaving a countercurrent cascade can at best be nearly at equilibrium with the feed stream. It is possible to have a richer extract by use of *reflux* at the product end of the cascade. This works like adding reflux to the rectifying section of a distillation column. An arrangement of countercurrent multistage extraction with reflux is schematically shown in Figure 8.14.

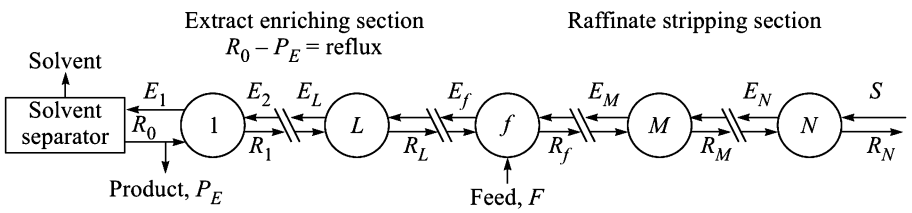


**Figure 8.14** Countercurrent multistage extraction with reflux.

### 8.4.6 The Minimum Solvent Rate

This is an important quantity often determined in connection with the design of an extraction equipment. An infinite number of contact stages (or an infinite height of a continuous contact

device such as a packed tower) is required to perform a given separation if the minimum solvent rate is used. If the Δ point for an infinite number of stages is known, the minimum solvent rate, $S_m$ (the suffix $m$ indicates 'minimum'), can be calculated by using the Lever Arm Rule.

$$\frac{S_m}{R_N} = \frac{\text{length of the line segment } R_N\Delta_m}{\text{length of the line segment } S\Delta_m} \tag{8.17}$$

To determine the minimum solvent rate $S_m$, the equilibrium curve is drawn on a suitable coordinate (for example, the right triangular coordinate) and the points $F$, $S$ and $R_N$ are located from the given compositions of the feed, the solvent and the raffinate. The points $R_N$ and $S$ are joined and produced. A tie line through the point $F$ and a few others below it are drawn. All these tie lines are extended to meet the line $R_NS$ produced at a number of Δ points. The meeting point *closest* to $S$ is $\Delta_m$ (see Figure 8.15) if the tie lines are inclined towards the point $S$ for the particular liquid–liquid system. There may be systems for which the tie lines slope away from the point $S$. In such a case, the extended tie lines intersect the line $SR_N$ produced on the left side of the diagram yielding a set of Δ points. The minimum solvent rate corresponds to the Δ point *farthest* from $S$ if it lies on the left or nearest to $S$ if it lies on the right. It may be noted that in many cases, the tie line passing through the feed point $F$ corresponds to the minimum solvent rate. This means that the pinch point lies on the feed end of the cascade. Determination of the minimum solvent rate is illustrated in Example 8.5.

Once the minimum solvent rate is found, the actual solvent rate is chosen on the basis of economic considerations. A higher solvent rate will no doubt reduce the fixed costs because less number of stages are required, but the operating cost will be higher primarily because of (i) handling a larger quantity of liquids, (ii) larger equipment size, and (iii) higher cost of separation of the solvent.

**EXAMPLE 8.5** (*Minimum solvent rate calculation*) Acetic acid ($C$) is to be extracted from a 45% aqueous solution using isopropyl ether ($B$) as the solvent at 20°C. The feed rate is 1500 kg/h and the raffinate must not contain more than 2.5% acid. If the solvent supplied to the extractor has 0.5% acetic acid in it, calculate the minimum solvent rate. The liquid–liquid equilibrium data at 20°C are given below.

| Water layer (*raffinate*), mass% | | | Ether layer (*extract*), mass% | | |
|---|---|---|---|---|---|
| *A* | *B* | *C* | *A* | *B* | *C* |
| 0.981 | 0.012 | 0.0069 | 0.005 | 0.993 | 0.0018 |
| 0.971 | 0.015 | 0.0141 | 0.007 | 0.989 | 0.0037 |
| 0.955 | 0.016 | 0.0289 | 0.008 | 0.984 | 0.0079 |
| 0.917 | 0.019 | 0.0642 | 0.01 | 0.971 | 0.0193 |
| 0.844 | 0.023 | 0.133 | 0.019 | 0.933 | 0.0482 |
| 0.711 | 0.034 | 0.255 | 0.039 | 0.847 | 0.114 |
| 0.589 | 0.044 | 0.367 | 0.069 | 0.715 | 0.216 |
| 0.451 | 0.106 | 0.443 | 0.108 | 0.581 | 0.311 |
| 0.371 | 0.165 | 0.464 | 0.151 | 0.487 | 0.362 |

### *Solution*

The equilibrium data are plotted on the right triangular coordinate to get the equilibrium curve (Figure 8.15). A set of tie lines are also drawn (these are the dotted lines in Figure 8.15). The

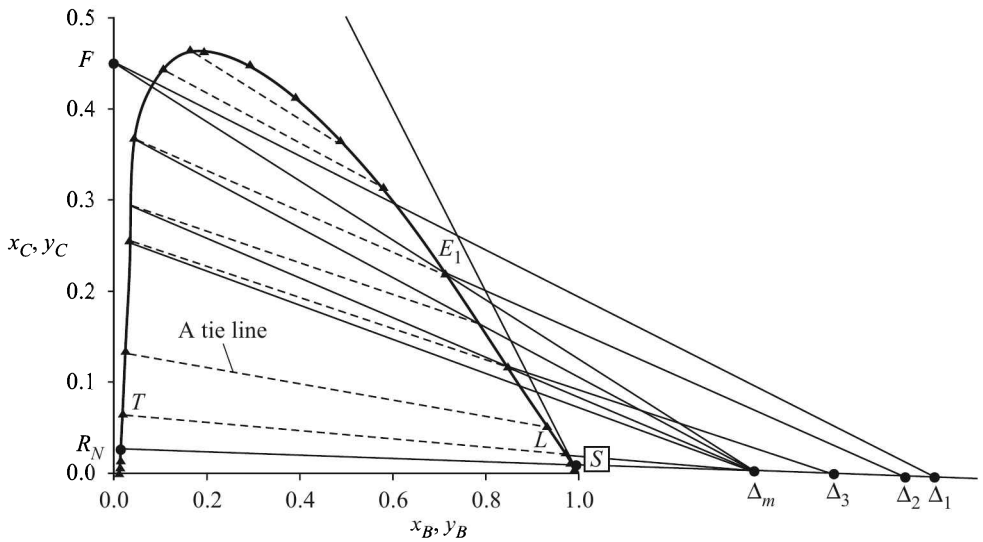


**Figure 8.15** Graphical determination of the minimum solvent rate ($\Delta_m$ is the difference point corresponding to the minimum solvent rate).

point $F$ (feed $F$ = 1500 kg/h; $x_{B,F}$ = 0, $x_{C,F}$ = 0.45) and the point $S$ (solvent; $y_{B,S}$ = 0.995, $y_{C,S}$ = 0.005) are located on the diagram. The raffinate has 2.5 mass% acid in it. The point $R_N$, representing the raffinate, is located on the equilibrium curve corresponding to $x_{C,RN}$ = 0.025. The points $R_N$ and $S$ are joined and the line $R_NS$ is extended.

A set of tie lines are drawn including one which, when extended, passes through the feed point $F$. The tie lines are found to slope towards the point $S$, and meet the extension of the line $R_NS$ at a number of $\Delta$ points. The $\Delta$ point ($\Delta_m$) nearest to $S$ corresponds to the minimum solvent rate, $S_m$. Here it happens for the tie line $TL$. We use Eq. (8.17) and material balances to calculate $S_m$.

The length segments $R_N\Delta_m$ (= 4.13 units) and $S\Delta_m$ (= 1.17 units) are measured from the diagram.

Then $$S_m/R_N = R_N\Delta_m/S\Delta_m = 4.13/1.17 = 3.53 \quad \text{(i)}$$

But $R_N$ is not known, and $S_m$ cannot be obtained from the above equation alone. It is necessary to obtain the concentration of the extract stream $E_1$, corresponding to the minimum solvent rate. The line $F\Delta_m$ intersects the extract arm of the equilibrium curve at $E_1$ and the concentration of the solute in it is found to be $y_1$ = 0.232. Now we can write down two more equations—the total material balance and the solute balance.

Total material balance: $$F + S_m = R_N + E_1 \Rightarrow 1500 + S_m = R_N + E_1 \quad \text{(ii)}$$

Solute balance: $$F(0.45) + S_m(0.005) = R_N(0.025) + E_1(0.232) \quad \text{(iii)}$$

Solving the above Eqs. (i), (ii) and (iii) we get, $S_m$ = $\boxed{1942.4 \text{ kg}}$ = minimum solvent rate.

## 8.5 LIQUID–LIQUID EXTRACTION EQUIPMENT

The liquid–liquid extraction operation involves intimate contact between two immiscible or partially miscible liquid phases compared to a gas or vapour and a liquid phase in gas absorption and distillation. A few major differences between these two important fluid–fluid contacting phenomena are discussed here.

(a) Dispersing one liquid in another one is more difficult than dispersing a gas in a liquid. When a gas bubbles through a liquid on a tray, the large difference in the densities of the phases and the small viscosity of the gas cause a high bubble rise velocity. This creates turbulence, rapid bubble breakage and coalescence and a large interfacial area of contact. But the shear stress required for the breakage of a drop is much higher than that for a bubble. Since the rise velocity of a drop in a liquid is rather small (because of small buoyancy), the required shear stress is not always generated by the motion of a drop alone.

(b) Unlike the gas–liquid system, a small density difference in a liquid–liquid system (besides a low interfacial tension) causes difficulty in separating a liquid–liquid dispersion into two clear liquid phases.

(c) A substantially higher mechanical energy input per unit volume of the medium is required for creating a liquid–liquid dispersion compared to that of a gas–liquid dispersion in a stirred vessel.

For the reasons mentioned above, the solvent extractors mostly have design and construction features much different from those of absorption and distillation columns. A classification of the more common extractors is given in Figure 8.16 (Reissinger and Schroter, 1978; Lo et al., 1983). Their construction and operation are briefly described below. Table 8.3 shows some important characteristics of common extractors.

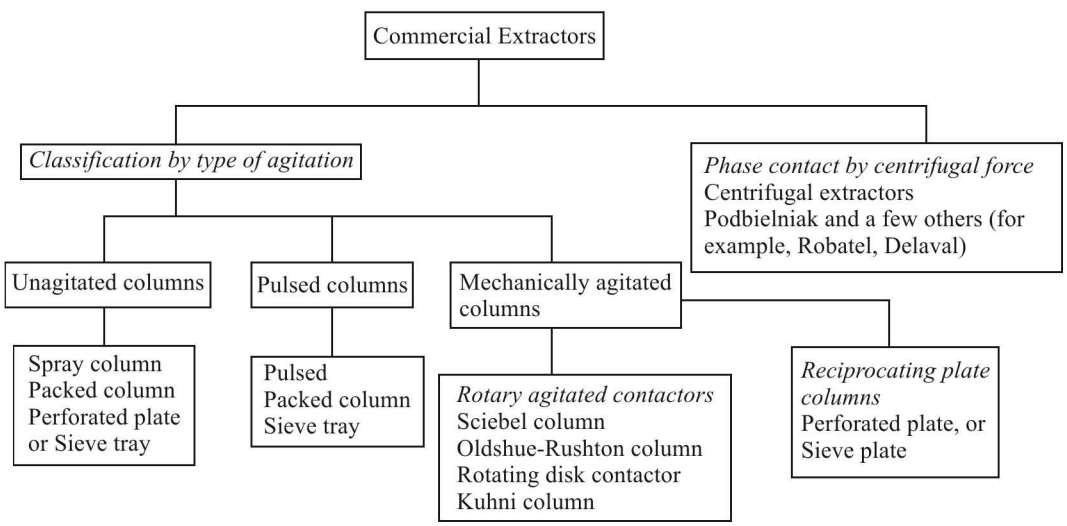


**Figure 8.16** Classification of commercial extractors.

**Table 8.3** Characteristics of the common extractors (Wankat, 1988)

| *Extractor class* | *Extractor type* | *Important characteristics* |
|---|---|---|
| Mixer-settlers | Agitated vessel<br>Static mixers | High stage efficiency, handles wide phase ratios and liquids with high viscosity, flexibility, reliable scale-up, not suitable for large number of theoretical stages. |
| Unagitated columns | Plate, packed and spray columns | Low capital cost, low operating/maintenance cost, simple construction, handles corrosive materials. |
| Pulsed columns | Perforated plate<br>Packed columns | Low HETS, high mass transfer coefficient, no internal moving parts. |
| Rotary agitated columns | RDC, Oldshue-Rushton, Scheibel, Kuhni extractor | Moderate to high capacity, reasonable HETS, accommodates many theoretical stages, moderate cost, low cost of operation and maintenance. |
| Reciprocating plate | Karr column | High loading, low HETS, high versatility and flexibility, simple construction, handles liquids with suspended solids. |
| Centrifugal | Podbielniak | Short contact time, suitable for unstable materials, small size (limited space requirement), handles easily emulsifiable and small $\Delta\rho$ systems. |

### 8.5.1 The Mixer-settler

This is the simplest and oldest type of extraction equipment. A stage of a mixer-settler cascade [Figure 8.7(a)] consists of a 'mixer' and a 'settler'.

#### *Mixer*

A common mixer is a vertical vessel in which one liquid phase is finely dispersed into the other by an agitator. The vessel is generally provided with four baffles to prevent the formation of vortex and also to create intense turbulence. The two liquid streams enter the mixer near the bottom. Sufficient time of contact is allowed for mass transfer from one phase to the other. This is the 'holding time'. The dispersion leaves the mixer at an upper level. A liquid–liquid mixing vessel is broadly similar to the one shown in Figure 5.15 except that it should be provided with two liquid inlets and one outlet for the dispersion.

A variety of impeller designs are available, for example, propeller, turbine, paddle, etc. A few common impeller types are shown in Figure 8.17. An impeller is usually positioned halfway the liquid depth. Power consumption or the rate of mechanical energy input depends upon a number of factors—the dimensions of the tank, the volume of dispersion, the impeller size and the design, the physical properties of the liquids (density, viscosity, interfacial tension), the ratio of the masses of the liquids in the tank ('phase ratio'), the degree of dispersion required, etc. Another important factor is, which phase is dispersed in the two-phase dispersion. If we start with pure $A$ (the carrier) and go on adding $B$ (the solvent) to it, we see that up to a certain mass per cent of $A$, it (i.e. the phase $A$) remains the continuous phase and $B$ remains in it in the form of droplets. But after a threshold phase ratio, *phase inversion* occurs when $B$ becomes the continuous phase while $A$ remains dispersed in it. The power consumption, hydrodynamics and mass transfer characteristics of a liquid–liquid dispersion greatly change when the inversion point is crossed.

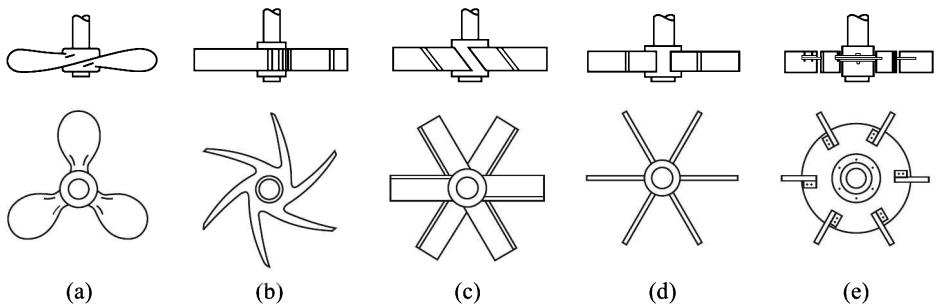


**Figure 8.17** A few common impeller types: (a) marine-type propeller, (b) centrifugal turbine, (c) pitched-blade turbine, (d) flat-blade paddle, and (e) flat-blade turbine (Treybal, *Mass Transfer Operations*, 1980).

As an alternative to mechanical agitation, dispersion of the phases in the mixing vessel may be achieved by pump circulation, inline non-mechanical mixing by using a jet or nozzle, a static mixer or even by air agitation. It is understandable that an intense agitation or mixing will produce very small droplets and will provide a high rate of mass transfer because of enhanced interfacial area and mass transfer coefficient. Even a stage-efficiency of nearly 100% may be achieved. But this benefit is partially overshadowed by the accompanying problem of separation of the dispersion in distinct phases in the settler. So, a balance between the two—namely creating a very fine dispersion to achieve a high stage-efficiency and the time and efficiency of settling—has to be reached. Drop-size in dispersions typically ranges between 0.1 and 1 mm.

### *Settler*

A dispersion leaves the mixer almost as an emulsion. This emulsion separates into two phases in the settler by coalescence of the fine droplets. A higher interfacial tension between the liquids enhances the coalescence rate; a lower value tends to produce a relatively stable emulsion that is difficult to separate into two phases. This factor also governs the selection of a solvent in liquid extraction. Phase separation becomes faster if the drop-size is larger, both the interfacial tension and the density difference between the liquids are higher, and the viscosity of the continuous phase is low. It may be noted that the drops are not of uniform size. There is always a 'distribution' of drop-size and 'drop-size' generally means the average size of the drops.

In order to improve the separation efficiency and to reduce the holding time and the volume of the settler, settling aids may be used. Tower packings, wire-mesh packings or oblique plate nests are effective settling aids. The surface of the settling aid is better wetted by the droplets or the dispersed phase. So, the impinging drops stick to the settling aid and then coalesce to form bigger drops that separate out of the dispersion.

A settling tank is a horizontal vessel provided with horizontal baffles (Figure 8.18). Typically, it has a length up to four times the diameter. For a system having $\Delta\rho \geq 0.1$ g/cm$^3$, a settler can separate about 300 gal/h of dispersion for 1 ft$^2$ of phase disengagement area (it is taken as $LD_t$, $L$ = vessel length, $D_t$ = diameter).

It will be pertinent to identify the advantages and disadvantages of the mixer–settler device. The *advantages* are: (a) high-stage efficiency (90% or more); (b) operational flexibility—a mixer-settler can handle difficult-to-disperse systems, like those having a large interfacial tension and/or

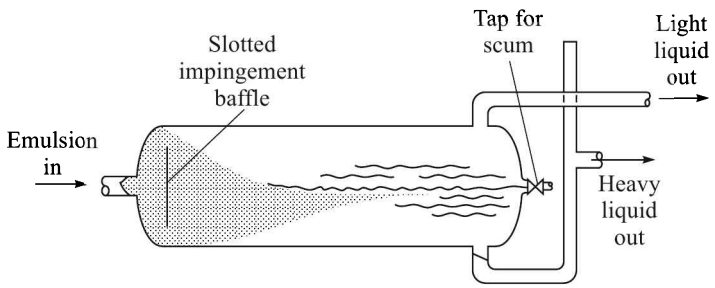


**Figure 8.18** Horizontal gravity-settling vessel for phase separation of liquid–liquid dispersion (Treybal, 1963).

large density difference; it can also operate with varying flow rates and phase ratios; (c) scale-up design of mixer-settlers can be done more reliably from laboratory test data; (d) it can handle high viscous liquids and also liquids having suspended solids. The *disadvantages* are: (a) considerable pump and piping cost (for this reason, the use of mixer-settlers is often limited to separation jobs requiring less than five theoretical stages); (b) a larger floor area is required; (c) more solvent cost, because a large volume of solvent is always retained in the settling vessels.

Mixer-settlers are widely used in wet metallurgical processes.

### *Tower extractor*

The tower extractor, developed by Lurgi Co., accommodates the mixers and settlers into a compact continuously operated unit as shown in Figure 8.19. The mixers of the cascade are placed outside the tower which has compartments for separating the phases. A drop coalescence zone is provided between a mixer and the corresponding settler.

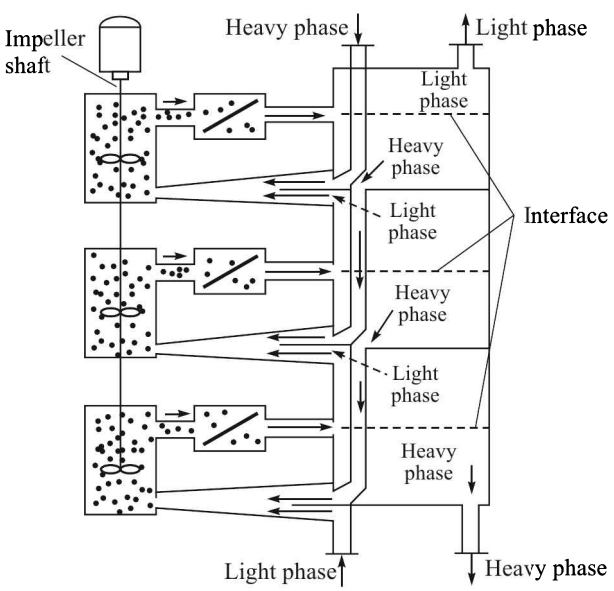


**Figure 8.19** Lurgi tower extractor (Humphrey et al., 1984).

### 8.5.2 Centrifugal Extractors

Centrifugal extractors are based on the use of 'centrifugal force' to cause countercurrent radial flow of the two liquid phases. Two common extractors of this kind are the 'Podbielniak extractor' and the 'Quadronic extractor'. The first one is the oldest centrifugal extractor put to commercial use.

The Podbielniak extractor (Figure 8.20) consists of a cylindrical drum rotating around a horizontal shaft. A number of concentric perforated cylindrical shells are fixed to the drum. The shaft is hollow except at the solid middle section. Both the heavy and the light phases enter through the shaft but from opposite directions. The heavy liquid is let out near the centre of the drum. The light liquid is channelled to the inner periphery of the rotating drum. Because of the density difference of the liquids, the heavy phase flows radially outwards and displaces the light liquid phase that flows radially inwards (i.e. in the opposite direction). The heavy liquid that reaches the periphery returns through the channels and leaves the unit through an annular region around the shaft. The light liquid collects near the centre and also leaves through an annular passage around the shaft. The Podbielniak extractor is *functionally* similar to a sieve tray column in which the trays, as if, form concentric perforated shells and countercurrent radial flow of the phases occurs.

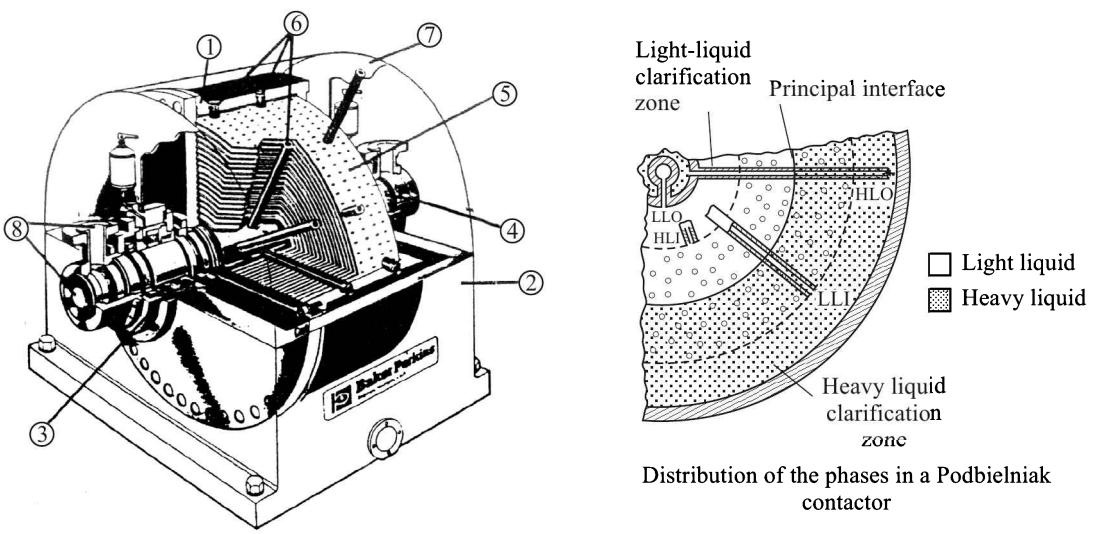


**Figure 8.20** The Podbielniak extractor: (1) rotor and shaft assembly, (2) base, (3) bearings, (4) multi-V-belt drive, (5) concentrically assembled perforated elements, (6) liquid distribution tubes, (7) tubular passage for cleaning and flushing, and (8) inlet and outlet ports.

The rotating drum, fixed to the trunnions carried on ball bearings, is provided with a drive pulley and is covered by an outer casing. Proper sealing arrangements are made at the contact between stationary and rotating parts. Drum diameters typically range from 1.5 to 3 ft, and the length may vary from 0.5 to 1 ft. The rotational speed generally varies from 2000 to 5000 rpm, and the contact time between the phases is only a few seconds. The extractor typically provides about 3 to 5 theoretical stages of contact and is suitable when the liquid density difference is $\Delta\rho \geq 0.05$ g/cm$^3$. The advantages of the Podbielniak extractor are: (a) very short contact time

between the phases (this makes the equipment suitable for unstable and sensitive materials like pharmaceuticals), (b) low space requirement, and (c) can handle liquids having a small difference in densities. The disadvantages are: (a) high capital cost, and (b) high maintenance cost.

### 8.5.3 Unagitated Extraction Columns

An extractor belonging to this category has a cylindrical shell with internals. The heavy liquid enters at the top and the light at the bottom through a distributor. One of the liquids remains dispersed in the other. Countercurrent flow of the liquids is driven by gravity and bouyancy forces. Drop breakage and coalescence occur all along the column and new interfacial area is created or the interfacial area is 'renewed'. The drop breakage and coalescence phenomena are enhanced in column extractors in which there is mechanical energy input by the motion of the internals (for example, a rotor) in the column.

Because the drops move under the effect of buoyancy only, the maximum allowable flow rates of the phases in a given column get limited by *flooding* of the tower. The mechanism of flooding is the same as that in a gas–liquid contactor. However, a small difference in the densities of the phases makes the flooding phenomenon more sensitive towards changes in flow rates.

Axial mixing of the phases occurs to some extent in the column. This reduces the theoretical rate of mass transfer in a countercurrent equipment. This phenomenon will be discussed later.

#### *Spray towers*

The spray tower is the simplest type of column extractor. It consists of a vertical cylinder with no internals, except spray nozzles at the top for the heavy liquid [Figure 8.21(a)]. Droplets of the dispersed phase form at the nozzles and move downwards through the light continuous phase. The 'throughputs' (i.e. the flow rates of the two phases through the column) greatly depend upon the density difference and viscosities of the phases. Alternatively, the heavy liquid

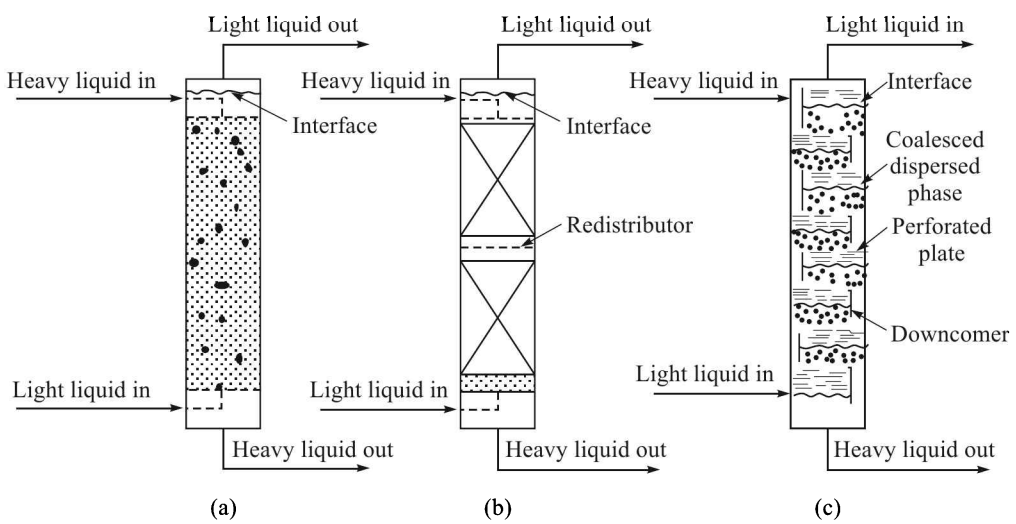


**Figure 8.21** Unagitated column extractors: (a) spray column, (b) packed column, and (c) perforated-plate (sieve-plate) column.

may be continuous and the light solvent may be distributed at the bottom through a perforated pipe to form droplets that will rise upwards. In this case, the drops finally separate out from the dispersion and coalesce into a clear liquid phase at the top above the level of inlet of the heavy phase from the bottom. Considerable 'axial mixing' (also called *backmixing*) occurs in the continuous phase, lowering the available driving force for mass transfer. The small rate of drop breakage and coalescence in the dispersion is also responsible for the low mass transfer rate. As a result, the spray tower does not normally give more than one to two theoretical stages.

Because of the simple construction and operation, spray columns are used for simpler operations like nutralization, washing, etc. But these are pretty common in 'direct contact' heating or cooling of liquids.

### *Packed extraction columns*

These are more efficient than the spray tower because the packing in the tower reduces backmixing and enhances drop breakage and coalescence (with frequent distortion) of the drops, thus providing a higher rate of mass transfer. The tower construction [Figure 8.21(b)] is similar to the packed gas absorber; it is provided with the packing support, the liquid distributor and redistributor, nozzles, etc. Similar packing materials are used (rings, saddles, modern high performance packings and structured packings). However, sufficient space must be provided above and below the liquid inlet nozzles for phase separation of the liquids. The separation efficiency of a packed column is strongly influenced by the wettability of the packing. If the packing is wetted by the dispersed phase, there will be more coalescence of the drops and hence the separation efficiency will be reduced. The 'height of a theoretical stage' (HETS) typically ranges between 0.6 and 1.6 m.

### *Sieve-tray columns*

The construction of a sieve-tray extraction column is similar to the one for gas absorption or distillation. The heavier phase flows across a perforated plate through which the lighter liquid rises as drops [Figure 8.21(c)]. The drops coalesce to form a phase-separated layer that remains in-between the dispersion and the immediate upper tray. This clear lighter phase then enters the upper tray as the dispersed drops. The heavier phase flowing down to the next lower tray is relieved of the lighter dispersed phase in the downcomer (this is comparable to the vapour–liquid disengagement in the downcomer in a distillation or gas absorption tower). The size of the perforations of a sieve tray may be ⅛ inch to ¼ inch. The tray spacing is considerably larger than that used in gas–liquid contacting. Weirs are *not needed* for a sieve tray extraction column.

### *Pulsed columns*

A periodic pulse to the accumulated heavier liquid at the bottom of a packed or a sieve tray is known to increase the mass transfer rate. A reciprocating unit imparts pulses (typically 1.5 to 4 hertz) to the lighter phases entering at the bottom. These are called 'pulsed extraction columns' and were quite popular at one time. Columns up to 1 m diameter are reportedly used in the nuclear industry (Ruthven, 1997).

Spray, packed and sieve-tray columns are non-proprietary columns (i.e. not covered by patents). But the class of column extractors with mechanical energy input are proprietary (patented) columns. The more important ones are described in the following section.